

Cover Page

FORSYT

Capstone Project Report

MID SEMESTER EVALUATION

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ABSTRACT

Geopolitical events such as wars, trade disputes, economic sanctions, diplomatic conflicts, and political instability significantly influence financial markets, supply chains, and investment decisions. Traditional geopolitical risk assessment approaches often rely on historical indices or descriptive reporting, limiting their ability to provide timely and actionable intelligence. Furthermore, existing systems generally focus on individual aspects of geopolitical analysis without integrating market intelligence, trade monitoring, and portfolio risk assessment into a unified decision-support platform.

Forsyt is an Artificial Intelligence-driven geopolitical intelligence platform designed to analyse geopolitical developments and assess their potential impact on Indian financial markets. The system collects and processes geopolitical news from multiple sources, applies Natural Language Processing techniques to extract meaningful information, and generates an India-centric Artificial Intelligence Geopolitical Risk (India AI-GPR) Index. The platform further integrates macroeconomic indicators, trade and supply chain intelligence, and portfolio exposure analysis to provide comprehensive decision support for investors, analysts, and policymakers.

The proposed system is developed using Python, React.js, PostgreSQL, Hugging Face Transformers, XGBoost, and SHAP, along with supporting technologies for data acquisition, processing, visualization, and deployment. The interactive dashboard presents geopolitical trends, macroeconomic indicators, market sentiment, portfolio risk assessments, and supply chain intelligence through an intuitive user interface.

Forsyt demonstrates the potential of combining Artificial Intelligence, Natural Language Processing, explainable machine learning, and interactive visualization to improve geopolitical intelligence and financial decision-making. The proposed platform provides a scalable and modular foundation for integrating real-time geopolitical analysis with investment risk assessment and supports future enhancements through additional analytical capabilities and data sources.

DECLARATION

We hereby declare that the Capstone Project entitled "Forsyt: A Geopolitical Risk Intelligence System for Indian Markets" is an authentic record of the work carried out by us in the Department of Computer Science and Engineering, Thapar Institute of Engineering and Technology, Patiala, under the guidance of Dr. Kapil Tomar and Dr. Jasmeet Singh during the Sixth Semester of the Bachelor of Engineering (Computer Engineering) programme.

We further declare that this work has not been submitted, either wholly or partially, for the award of any other degree or diploma at any institution.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
AI-GPR	Artificial Intelligence Geopolitical Risk
API	Application Programming Interface
CSV	Comma-Separated Values
EPU	Economic Policy Uncertainty
FII	Foreign Institutional Investor
GDP	Gross Domestic Product
GPR	Geopolitical Risk
JSON	JavaScript Object Notation
ML	Machine Learning
NLP	Natural Language Processing
REST	Representational State Transfer
RSS	Really Simple Syndication
SHAP	SHapley Additive exPlanations
UI	User Interface
USD	United States Dollar
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Project Overview

Geopolitical developments have become one of the most significant external factors influencing the stability and performance of modern financial markets. Events such as international conflicts, trade disputes, economic sanctions, diplomatic tensions, disruptions to strategic shipping routes, and changes in global policy often create uncertainty that extends well beyond the political domain. These developments influence commodity prices, exchange rates, international trade, investor sentiment, and equity markets, making geopolitical risk an important consideration in financial decision-making. As financial markets become increasingly interconnected, the ability to systematically analyse geopolitical events and understand their economic implications has become essential for investors, businesses, researchers, and policymakers.

The Indian economy is particularly sensitive to geopolitical uncertainty due to its expanding participation in global trade, dependence on critical imports such as crude oil and semiconductors, growing manufacturing sector, and rapidly increasing retail investment ecosystem. Disruptions occurring outside India's borders frequently influence domestic inflation, logistics costs, industrial production, and market volatility. At the same time, the number of retail investors participating in Indian capital markets has grown significantly over the past few years, resulting in an increased demand for reliable, timely, and data-driven market intelligence. While financial information is readily available through numerous online platforms, geopolitical information remains fragmented across multiple news sources and often requires substantial manual effort to interpret. Consequently, investors and analysts are frequently required to correlate geopolitical developments with financial indicators using subjective judgement rather than structured analytical evidence.

Existing geopolitical monitoring frameworks primarily focus on measuring global geopolitical uncertainty using international news coverage and generalized risk indicators. Although these systems have contributed significantly to geopolitical risk measurement, most are designed from a global or Western perspective and therefore provide limited insight into India's unique economic dependencies and strategic priorities. Furthermore, many existing solutions emphasize monitoring rather than actionable analysis. They report geopolitical events but seldom quantify their direct implications for Indian financial markets, investment portfolios, or supply chain resilience. Commercial geopolitical intelligence platforms offer more advanced capabilities; however, their proprietary nature often limits accessibility for academic researchers, students, and individual investors. These limitations highlight the need for

a transparent and India-centric analytical framework capable of transforming geopolitical information into practical decision-support intelligence.

To address this requirement, **Forsyt** has been proposed as an Artificial Intelligence-driven geopolitical risk intelligence platform specifically designed for the Indian financial ecosystem. The project aims to transform unstructured geopolitical news into structured analytical information that can be used to evaluate geopolitical risk and its potential economic consequences. Rather than functioning solely as a news aggregation platform or a predictive market model, Forsyt integrates data acquisition, natural language processing, quantitative risk assessment, machine learning, explainable artificial intelligence, and interactive visualization within a unified software system. This integrated approach enables users to move beyond qualitative interpretation of news by providing measurable indicators and analytical insights that support evidence-based decision-making.

The foundation of Forsyt is an automated pipeline that continuously collects geopolitical news from multiple curated sources and converts unstructured textual information into structured event data. Through Natural Language Processing techniques, relevant entities, geopolitical events, affected regions, industrial sectors, and contextual information are identified and organized into a standardized representation suitable for computational analysis. This structured information forms the basis for generating the **India Artificial Intelligence Geopolitical Risk Index (India AI-GPR)**, a quantitative indicator designed to measure the level of geopolitical risk affecting the Indian economic environment. Unlike conventional geopolitical indices that provide broad measures of global uncertainty, the proposed index emphasizes geopolitical developments that possess direct or indirect relevance to India's financial and economic interests.

Building upon the India AI-GPR Index, Forsyt provides multiple analytical modules that collectively support different aspects of geopolitical decision-making. The **News Intelligence System** enables users to explore structured geopolitical events extracted from news articles. The **Market Intelligence Module** investigates the relationship between geopolitical developments and volatility in the Nifty 50 index using machine learning techniques. The **Portfolio Risk Advisor** estimates sector-wise geopolitical exposure to assist investors in understanding portfolio sensitivity during periods of elevated geopolitical uncertainty. In addition, the **Supply Chain Risk Assessment Module** analyses risks associated with strategically important trade routes and international logistics corridors. Since all analytical modules operate on a common geopolitical intelligence framework, information generated by one component can be consistently utilized by the others, resulting in an integrated rather than fragmented analytical environment.

An important design objective of the proposed platform is to ensure that generated insights remain interpretable to end users. Artificial Intelligence models are increasingly capable of producing highly accurate predictions; however, their practical adoption is often limited when users cannot understand the reasoning behind model outputs. Forsyt therefore incorporates explainable analytical techniques to complement predictive models with meaningful interpretations of the factors contributing to generated risk assessments. This approach enables users to evaluate not only the predicted outcome but also the geopolitical drivers responsible for changes in market conditions or risk levels, thereby improving transparency and supporting more informed decision-making.

The development of Forsyt also demonstrates the integration of multiple engineering disciplines within a single software system. The project combines concepts from Natural Language Processing, Machine Learning, Database Management Systems, Software Engineering, Data Analytics, Web Technologies, and Financial Engineering to develop a scalable and modular analytical platform. Rather than addressing these technologies independently, the project illustrates how they can be integrated to solve a practical problem involving large-scale data processing, intelligent information extraction, quantitative modelling, and interactive visualization. The modular architecture further facilitates maintainability, future expansion, and independent development of analytical components while ensuring that the overall system remains cohesive.

Overall, Forsyt is envisioned as a comprehensive geopolitical risk intelligence platform that bridges the gap between unstructured geopolitical information and structured financial decision support. By integrating automated news analysis, India-centric geopolitical risk quantification, predictive analytics, explainable artificial intelligence, and interactive visualization into a unified framework, the project seeks to provide timely, transparent, and actionable geopolitical intelligence for the Indian financial ecosystem. The subsequent sections of this report discuss the motivation behind the project, analyse existing research, identify research gaps, describe the adopted methodology, and present the design and implementation of the proposed system.

1.2 Need Analysis

The increasing influence of geopolitical events on financial markets has created a growing demand for systems capable of translating complex geopolitical developments into actionable financial intelligence. International conflicts, trade restrictions, economic sanctions, diplomatic negotiations, disruptions to global supply chains, and fluctuations in energy markets frequently affect investment decisions, market volatility, and economic stability. While these developments receive extensive media coverage, the information is typically dispersed across numerous news platforms and presented in an unstructured format. Consequently, investors, analysts, and decision-makers often face

significant difficulty in identifying which events are genuinely relevant to the Indian financial ecosystem and assessing their potential economic impact.

The Indian financial market has witnessed substantial growth in retail investor participation over the past decade, resulting in a larger and more diverse investor community that increasingly relies on publicly available information for financial decision-making. At the same time, India's expanding integration with global trade and international supply chains has made domestic markets more susceptible to geopolitical developments occurring beyond national borders. Events affecting crude oil supply, semiconductor availability, maritime transportation, international trade agreements, and regional conflicts often influence market sentiment and sector-specific performance within India. However, there is currently no widely accessible platform that systematically monitors such developments, evaluates their relevance to India, and presents the resulting intelligence in a structured and interpretable manner.

Existing information sources generally address only a portion of the overall problem. Financial news portals primarily focus on reporting market movements, whereas international news organizations emphasize geopolitical developments without explicitly analysing their implications for Indian financial markets. Geopolitical risk indices developed by research institutions provide valuable measures of global uncertainty but are not specifically designed to capture India's unique economic priorities or translate geopolitical events into practical investment intelligence. As a result, users are often required to manually collect information from multiple platforms, interpret geopolitical developments independently, and estimate their possible influence on financial markets using subjective judgement. This fragmented approach is both time-consuming and susceptible to inconsistencies.

The need for Forsyt therefore arises from the requirement for an integrated platform that combines automated news monitoring, geopolitical event extraction, quantitative risk assessment, predictive analytics, and interactive visualization within a unified framework. By consolidating these capabilities into a single system, the proposed platform seeks to reduce information fragmentation, improve the accessibility of geopolitical intelligence, and support evidence-based financial decision-making. The project further addresses the growing requirement for transparent Artificial Intelligence systems by complementing predictive models with interpretable analytical outputs, enabling users to understand not only the generated predictions but also the geopolitical factors contributing to them.

In addition to supporting investors and financial analysts, the proposed platform is expected to benefit researchers, policymakers, and supply chain managers who require timely geopolitical intelligence for strategic planning and risk assessment. Through an

India-centric approach to geopolitical risk analysis, Forsyt aims to bridge the gap between raw geopolitical information and meaningful decision support, thereby contributing to a more informed and resilient financial ecosystem.

1.3 Research Gaps

The review of the existing literature and currently available geopolitical risk assessment approaches highlights several limitations that restrict their applicability for comprehensive geopolitical intelligence within the Indian financial ecosystem. The major research gaps identified during the study are summarized in **Table 1.1**.

Research Gap	Impact on Existing Approaches
Existing geopolitical risk indices primarily measure global geopolitical uncertainty and are not specifically designed for the Indian financial ecosystem.	Limited relevance for analysing the impact of geopolitical events on Indian financial markets.
Most studies focus on either geopolitical risk assessment or financial market prediction independently rather than integrating both into a unified analytical framework.	Users are required to manually interpret geopolitical developments before making financial decisions.
Existing systems largely emphasize monitoring and reporting geopolitical events instead of providing actionable decision-support insights.	Reduces the practical usability of geopolitical intelligence for investors and analysts.
Limited emphasis has been placed on explainability within AI-based geopolitical forecasting models.	Lack of transparency reduces user confidence in analytical predictions.
Existing solutions generally address isolated analytical tasks instead of integrating news intelligence, geopolitical risk quantification, market analysis, and visualization within a single platform.	Results in fragmented workflows and increased dependence on multiple independent tools.

The identified research gaps collectively establish the need for an integrated geopolitical risk intelligence platform capable of transforming unstructured geopolitical information into structured analytical insights. **Forsyt** addresses these limitations by combining

automated news intelligence, India AI-GPR Index generation, predictive market analysis, explainable artificial intelligence, portfolio risk assessment, and supply chain risk analysis within a unified decision-support framework. The manner in which the proposed system builds upon existing research while addressing these identified gaps is discussed in detail in **Chapter 2: Literature Survey**.

1.4 Problem Definition and Scope

1.4.1 Problem Definition

Geopolitical events such as armed conflicts, economic sanctions, diplomatic disputes, trade restrictions, and disruptions to international supply chains significantly influence financial markets and economic activity. Although large volumes of geopolitical information are published daily through news portals and media outlets, this information remains largely unstructured, fragmented, and difficult to interpret from a financial perspective. Investors, researchers, policymakers, and businesses often rely on manual analysis to understand the potential implications of these events on the Indian economy, making the decision-making process time-consuming, inconsistent, and highly dependent on individual interpretation.

Existing geopolitical risk indices provide valuable measures of global uncertainty; however, they are not specifically designed to represent the Indian financial ecosystem or quantify the direct influence of geopolitical developments on Indian markets. Similarly, financial analysis platforms primarily focus on market performance without integrating structured geopolitical intelligence, while news platforms provide information without systematically translating it into measurable financial risk indicators. This disconnect between geopolitical information and financial analysis creates a significant challenge for stakeholders seeking timely and evidence-based decision support.

The primary problem addressed by Forsyt is the absence of an integrated, India-centric platform capable of automatically collecting geopolitical news, extracting meaningful events, quantifying geopolitical risk, and presenting actionable analytical insights relevant to financial decision-making. The project seeks to reduce the dependence on manual interpretation by transforming unstructured geopolitical information into structured intelligence that can be analysed using Artificial Intelligence and Machine Learning techniques. Through this approach, Forsyt aims to provide a unified framework that supports geopolitical monitoring, quantitative risk assessment, market analysis, and decision support within a single software platform.

1.4.2 Scope of the Project

The scope of Forsyt is focused on developing an integrated geopolitical risk intelligence platform for the Indian financial ecosystem. The project encompasses the complete analytical workflow beginning with the acquisition of geopolitical news from selected

information sources and extending to the generation of structured intelligence for decision support. The system includes automated news processing, geopolitical event extraction, India AI-GPR Index generation, predictive analysis of Nifty 50 market volatility, portfolio risk assessment, supply chain risk analysis, and visualization of analytical outputs through an interactive dashboard.

The project is primarily intended as a decision-support system rather than an autonomous financial advisory platform. Its objective is to assist users in understanding the potential implications of geopolitical developments by providing structured information, quantitative risk indicators, and interpretable analytical outputs. The generated predictions and risk assessments are designed to complement human judgement and should not be considered direct investment recommendations.

The current implementation is limited to geopolitical events relevant to the Indian economic environment and evaluates their influence using the datasets, analytical models, and validation framework adopted during the project. The scope does not include automated financial trading, personalized investment management, or prediction of individual stock prices. Instead, the platform concentrates on analysing broader geopolitical trends, estimating their potential impact on market behaviour, and providing an integrated environment for geopolitical intelligence and financial risk assessment.

Furthermore, the modular architecture adopted for Forsyt enables future enhancements without requiring significant structural modifications to the existing system. Additional analytical modules, alternative data sources, and improved predictive models can be incorporated as the platform evolves, allowing the project to remain extensible while maintaining the integrity of its core geopolitical intelligence framework.

1.5 Assumptions and Constraints

The development of **Forsyt** is based on several assumptions regarding data availability, system usage, and operational conditions. Additionally, certain constraints influence the scope and implementation of the proposed system. These assumptions and constraints are presented below.

1.5.1 Assumptions

The successful operation of the proposed system is based on the following assumptions.

Table 1.5 Assumptions of Forsyt

S. No.	Assumption
1	Reliable geopolitical news data is available from the configured news sources throughout the operation of the system.
2	Financial datasets and macroeconomic indicators used for analysis are assumed to be accurate, complete, and periodically updated.
3	Users access the platform through a stable internet connection using a modern web browser compatible with the web application.
4	The AI and NLP models are assumed to be trained using representative datasets capable of producing meaningful analytical results.
5	Users possess basic knowledge of financial markets and geopolitical concepts to correctly interpret the analytical outputs generated by the platform.
6	External libraries, APIs, and open-source frameworks used in the project remain available and function as expected during system execution.

1.5.2 Constraints

The implementation of Forsyt is subject to the following constraints.

Table 1.6 Constraints of Forsyt

S. No.	Constraint
1	The accuracy of analytical outputs depends on the quality, completeness, and timeliness of the collected geopolitical news data.
2	The current implementation primarily focuses on the Indian financial ecosystem and selected global geopolitical events relevant to India.
3	Real-time analysis is dependent on the availability and accessibility of external news sources and financial datasets.
4	Predictive analytical models provide decision support and should not be interpreted as guaranteed financial forecasts.
5	System performance may vary depending on the computational resources available on the deployment environment.

6	Future geopolitical developments and unforeseen global events may influence analytical accuracy beyond the scope of the trained models.
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Discussion

The assumptions and constraints outlined above define the operational environment within which Forsyt has been designed and implemented. While the proposed system provides AI-driven geopolitical intelligence and financial analytics, its effectiveness depends on the availability of reliable data sources, computational resources, and appropriate user interpretation. These factors have been considered during the system design to ensure a practical, scalable, and maintainable solution.

1.6 Standards

The development of **Forsyt** follows standard software engineering and documentation practices to ensure consistency, maintainability, and reproducibility throughout the project lifecycle. The standards and conventions adopted during the project are summarized in **Table 1.2**.

Area	Standard / Practice Followed	Purpose
Report Documentation	TIET Report Formatting Guidelines	Ensures uniform structure, formatting, and presentation of the technical report.
Referencing	IEEE Reference Style	Maintains consistency in citations and bibliography.
Software Modelling	Unified Modeling Language (UML)	Represents the structural and behavioural aspects of the system using standardized diagrams.
Version Control	Git & GitHub	Supports collaborative development, version management, and source code tracking.
Programming Practices	Modular Software Development	Promotes maintainability, scalability, and easier integration of project components.
Model Evaluation	Standard Evaluation Metrics adopted during implementation	Provides objective assessment of the developed NLP and machine learning models.

The adoption of these standards ensures that the proposed system is developed using recognized software engineering practices while maintaining consistency in implementation, documentation, evaluation, and collaborative development. Furthermore, the use of UML diagrams and standardized documentation improves the readability and maintainability of the project, while version control facilitates efficient project management and future enhancements.

1.7 Objectives

1.7.1 Primary Objective

The primary objective of Forsyt is to develop an Artificial Intelligence-driven geopolitical risk intelligence platform that systematically transforms unstructured geopolitical news into structured analytical intelligence for the Indian financial ecosystem. The platform aims to automate the collection, processing, analysis, and visualization of geopolitical information to support evidence-based decision-making through quantitative risk assessment, predictive analytics, and explainable insights.

1.7.2 Specific Objectives

To achieve the primary objective, the following specific objectives have been identified:

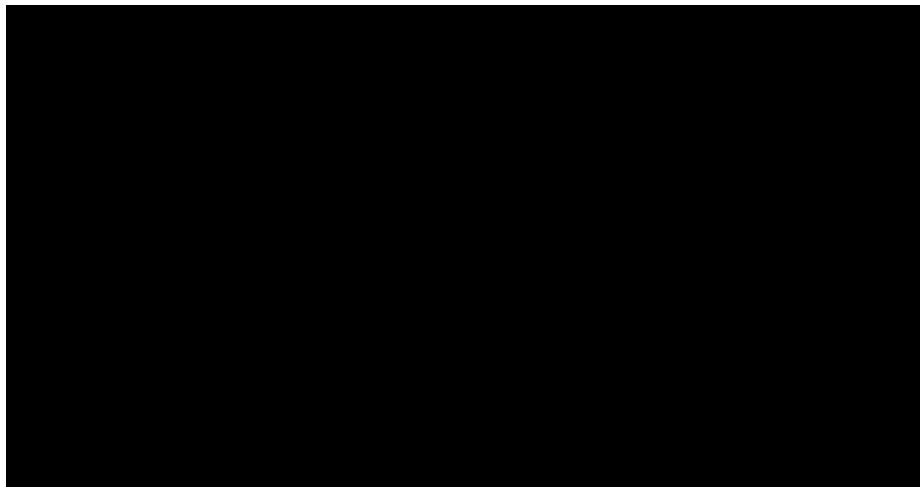
1. **To develop an automated news intelligence pipeline** capable of collecting and processing geopolitical news from selected sources and converting unstructured textual information into structured event data suitable for further analysis.
2. **To design and implement the India AI-GPR Index** by transforming extracted geopolitical events into a quantitative indicator that reflects geopolitical risk relevant to the Indian economic environment.
3. **To analyse the relationship between geopolitical risk and the Indian equity market** by utilizing the generated geopolitical risk index as an input for predicting volatility trends in the Nifty 50 index.
4. **To develop a Portfolio Risk Advisor** that evaluates the geopolitical exposure of investment portfolios by analysing sector-wise sensitivity to evolving geopolitical conditions.
5. **To design a Supply Chain Risk Assessment Module** capable of identifying and assessing geopolitical risks associated with important trade routes and logistics corridors relevant to India's economic activities.

6. **To incorporate Explainable Artificial Intelligence (XAI) techniques** that improve the transparency of predictive models by providing interpretable explanations for generated outputs and risk assessments.
7. **To develop an interactive visualization dashboard** that presents geopolitical events, analytical outputs, and risk indicators in a structured and user-friendly manner, enabling users to efficiently monitor and interpret geopolitical developments.
8. **To integrate all analytical components into a unified software platform** that minimizes information fragmentation and provides a comprehensive environment for geopolitical intelligence and financial decision support.

1.8 Methodology

The development of **Forsyt** follows a systematic methodology that converts unstructured geopolitical news into structured analytical intelligence through a sequence of interconnected processing stages. The methodology is modular in nature, allowing each phase to perform a dedicated function while collectively contributing to the overall objective of geopolitical risk assessment and financial decision support. The overall workflow adopted in the project is illustrated in **Figure 1.1**.

Figure 1.1 Overall Methodology of the Forsyt Platform



The methodology adopted for the development of Forsyt consists of the following phases:

Phase 1: Geopolitical News Collection

The process begins with the collection of geopolitical news from selected sources. News articles related to international conflicts, diplomatic developments, trade policies, economic sanctions, maritime disruptions, and other geopolitically significant events are gathered to create the primary dataset for analysis.

Phase 2: Data Pre-processing and NLP Pipeline

The collected news articles undergo pre-processing to remove inconsistencies and prepare the data for analysis. The Natural Language Processing (NLP) pipeline then extracts relevant geopolitical entities, events, locations, organizations, and contextual information, converting unstructured textual data into structured event information.

Phase 3: India AI-GPR Index Generation

The structured geopolitical events generated by the NLP pipeline are utilized to construct the **India Artificial Intelligence Geopolitical Risk Index (India AI-GPR Index)**. The index provides a quantitative representation of geopolitical risk relevant to the Indian financial ecosystem and serves as the core analytical output of the platform.

Phase 4: Pipeline Validation and Performance Evaluation

The generated outputs are validated using the evaluation methodology adopted for the project. The validation process assesses the effectiveness of the developed NLP pipeline and ensures that the generated geopolitical intelligence is consistent with the selected reference dataset.

Phase 5: Feature Engineering

The generated geopolitical risk indicators and structured event information are transformed into features suitable for downstream analytical tasks. This stage prepares the processed data for machine learning models used within the platform.

Phase 6: Machine Learning Model Development

Machine learning techniques are employed to analyse the relationship between geopolitical developments and financial market behaviour. The developed models utilize the processed features to support market analysis and other predictive components incorporated within Forsyt.

Phase 7: Dashboard Development and Visualization

The outputs generated by the analytical modules are presented through an interactive dashboard. The dashboard provides users with access to geopolitical events, India AI-GPR trends, analytical results, and visualizations within a unified interface, thereby facilitating efficient interpretation and decision support.

Phase 8: System Testing and Final Validation

The completed platform undergoes system testing to verify the correct integration of all software modules. This phase ensures that the individual components operate cohesively and that the developed platform performs in accordance with the intended project objectives.

The adopted methodology establishes a structured workflow that integrates news acquisition, Natural Language Processing, geopolitical risk quantification, machine learning, validation, and visualization into a unified software system. This modular approach improves maintainability, supports future enhancements, and enables the individual analytical components to operate together while contributing to the overall objective of developing an India-centric geopolitical risk intelligence platform.

1.9 Project Outcomes

The successful completion of **Forsyt** is expected to achieve the following outcomes:

Technical Outcomes

- Development of an **India-centric Geopolitical Risk Intelligence Platform** for monitoring and analysing geopolitical events affecting the Indian financial ecosystem.
- Establishment of an **automated News Intelligence System** capable of transforming unstructured geopolitical news into structured event data through Natural Language Processing.
- Generation of the **India Artificial Intelligence Geopolitical Risk Index (India AI-GPR Index)** to quantitatively represent geopolitical risk relevant to India.
- Development of a **Nifty 50 Volatility Prediction Module** to analyse the relationship between geopolitical risk and market behaviour using machine learning techniques.
- Implementation of a **Portfolio Risk Advisor** that evaluates sector-wise geopolitical exposure and assists in portfolio risk assessment.

- Development of a **Supply Chain Risk Assessment Module** to identify potential geopolitical disruptions affecting major trade routes and logistics corridors.
- Integration of **Explainable Artificial Intelligence (XAI)** to improve transparency by providing interpretable explanations for analytical outputs and model predictions.
- Development of an **interactive dashboard** that consolidates geopolitical events, India AI-GPR trends, predictive analytics, and risk assessments into a unified user interface.

Engineering Outcomes

- Demonstration of the practical integration of **Artificial Intelligence, Natural Language Processing, Machine Learning, Data Analytics, and Software Engineering** within a single modular software platform.
- Development of a scalable and modular architecture that supports future integration of additional datasets, analytical models, and application modules.
- Validation of the developed analytical pipeline using the evaluation methodology adopted during the project, ensuring that generated outputs are assessed against the selected reference dataset.

Academic Outcomes

- Application of interdisciplinary engineering concepts to solve a real-world problem involving geopolitical intelligence and financial decision support.
- Development of a reusable framework that can support future academic research in geopolitical risk analysis, financial forecasting, and intelligent decision-support systems.

1.10 Novelty

The novelty of **Forsyt** lies in the integration of multiple analytical capabilities into a single India-centric geopolitical risk intelligence platform. While existing research primarily focuses on individual aspects such as geopolitical risk measurement, financial forecasting, or natural language processing, Forsyt combines these capabilities within a unified framework to provide structured geopolitical intelligence and financial decision support.

The distinguishing features of the proposed system are summarized in **Table 1.3**.

Table 1.3 Novel Features of Forsyt

Existing Approaches	Forsyt
Primarily designed for measuring global geopolitical uncertainty.	Focuses on India-centric geopolitical risk assessment relevant to the Indian financial ecosystem.
News monitoring and financial analysis are generally performed using separate systems.	Integrates news intelligence, geopolitical risk assessment, financial analytics, and visualization within a single platform.
Most approaches provide descriptive analysis of geopolitical events.	Generates structured analytical insights through the India AI-GPR Index and downstream analytical modules.
Limited transparency in AI-based analytical models.	Incorporates Explainable AI (XAI) to improve interpretability of analytical outputs.
Existing solutions often address a single application domain.	Supports multiple application areas, including market analysis, portfolio risk assessment, and supply chain risk assessment within one framework.
Many platforms require manual interpretation of dispersed information.	Automates the transformation of unstructured geopolitical news into structured decision-support intelligence through an integrated NLP pipeline.

The novelty of Forsyt does not lie in introducing entirely new algorithms or replacing existing geopolitical risk models. Instead, it lies in the integration of established Artificial Intelligence, Natural Language Processing, Machine Learning, and data analytics techniques into a unified, India-centric software platform that transforms fragmented geopolitical information into structured analytical intelligence. By combining multiple analytical modules within a common framework, the proposed system provides a comprehensive approach to geopolitical risk assessment and financial decision support while addressing the practical limitations identified during the literature review.

2. REQUIREMENT ANALYSIS

2.1 Literature Survey

2.1.1 Theory Associated with Problem Area

The increasing frequency of geopolitical conflicts, trade disputes, economic sanctions, diplomatic tensions, and supply chain disruptions has significantly influenced global financial markets. These events generate large volumes of unstructured information through news articles, government releases, and media reports, making manual analysis both time-consuming and inconsistent. Consequently, intelligent analytical systems capable of extracting meaningful information from geopolitical news have become increasingly important for supporting financial decision-making.

Forsyt addresses this challenge by integrating geopolitical news analysis with Artificial Intelligence, Natural Language Processing, and financial analytics to transform unstructured information into structured decision-support intelligence. The theoretical concepts associated with this problem area are discussed in the following subsections.

2.1.1.1 Geopolitical Risk

Geopolitical risk refers to the uncertainty arising from political, economic, military, diplomatic, or social events that influence national economies, international trade, and financial markets. Events such as armed conflicts, sanctions, elections, policy changes, and disruptions to global supply chains often affect investor confidence and market behaviour.

Unlike conventional financial indicators, geopolitical events are qualitative in nature and difficult to quantify directly. Their impact varies depending on geographical location, duration, severity, and the industries affected. Consequently, continuous monitoring and systematic assessment of geopolitical developments have become essential for governments, businesses, and investors.

2.1.1.2 News Intelligence

News intelligence is the process of collecting, analysing, and interpreting information from large volumes of news articles to identify events, trends, and emerging risks. Instead of manually reading thousands of articles, intelligent news analysis systems automatically process textual information and identify relevant developments that require attention.

In geopolitical intelligence platforms, news serves as the primary source of information because major international events are first reported through media organizations before influencing economic indicators and financial markets. Automated news intelligence enables timely identification of geopolitical developments and supports faster analytical decision-making.

2.1.1.3 Natural Language Processing

Natural Language Processing (NLP) enables computers to understand, process, and analyse human language. Since geopolitical information is predominantly available as unstructured text, NLP techniques are employed to convert textual information into structured data suitable for computational analysis.

Typical NLP tasks include text preprocessing, tokenization, named entity recognition, event extraction, and relationship identification. These techniques enable intelligent systems to identify countries, organizations, political leaders, military activities, economic events, and diplomatic actions from news articles. The structured information generated through NLP forms the analytical foundation for subsequent risk assessment and predictive modelling.

2.1.1.4 Financial Market Analysis

Financial markets respond rapidly to geopolitical developments. Events such as military conflicts, trade restrictions, economic sanctions, and diplomatic negotiations often influence investor sentiment, commodity prices, stock market volatility, and capital flows.

Analysing the relationship between geopolitical events and financial markets enables investors and policymakers to better understand market behaviour during periods of uncertainty. Integrating geopolitical intelligence with financial analytics therefore improves situational awareness and supports more informed decision-making.

2.1.1.5 Explainable Artificial Intelligence

Explainable Artificial Intelligence (XAI) focuses on making AI-generated predictions understandable and transparent. Instead of presenting only predictive results, explainable models provide insights into the factors influencing their decisions.

In financial and geopolitical applications, explainability is particularly important because users often require justification before relying on analytical recommendations. Transparent analytical outputs improve user confidence and facilitate informed interpretation of generated insights.

2.1.1.6 Decision Support Systems

A Decision Support System (DSS) is a computer-based system designed to assist users in analysing complex problems by combining data, analytical models, and visualization techniques. Rather than replacing human judgment, a DSS organizes large volumes of information and presents meaningful insights to support effective decision-making.

In the context of geopolitical intelligence, decision-support systems integrate information from multiple analytical modules to provide a comprehensive understanding of geopolitical developments and their potential impact on financial markets.

Summary of Theoretical Concepts

Table 2.1 Theoretical Concepts Associated with the Problem Area

Concept	Role in Forsyt
Geopolitical Risk	Represents uncertainty arising from geopolitical developments affecting financial markets.
News Intelligence	Collects and analyses geopolitical news to identify significant events.
Natural Language Processing	Converts unstructured textual news into structured analytical information.
Financial Market Analysis	Studies the relationship between geopolitical events and market behaviour.
Explainable Artificial Intelligence	Improves transparency and interpretability of analytical outputs.
Decision Support Systems	Integrates analytical modules to support informed decision-making.

The above theoretical concepts collectively establish the foundation for the development of Forsyt. By integrating geopolitical news analysis, Natural Language Processing, financial analytics, and explainable Artificial Intelligence, the proposed platform transforms unstructured geopolitical information into structured analytical intelligence capable of supporting informed decision-making within the Indian financial ecosystem.

2.1.2 Existing Systems and Solutions

The increasing importance of geopolitical intelligence has encouraged researchers and organizations to develop systems capable of measuring geopolitical uncertainty, monitoring global events, and analysing their impact on financial markets. These systems differ in their objectives and methodologies. Some focus on quantifying geopolitical risk through index construction, while others specialize in news monitoring, financial prediction, or market intelligence. The major existing systems relevant to the proposed Forsyt platform are discussed below.

2.1.2.1 Geopolitical Risk (GPR) Index

One of the most widely accepted approaches for measuring geopolitical uncertainty is the **Geopolitical Risk (GPR) Index**, introduced by **Caldara and Iacoviello** in their research *Measuring Geopolitical Risk* [1]. The index quantifies geopolitical risk by analysing the frequency of newspaper articles related to geopolitical tensions, wars, terrorist activities, military conflicts, and diplomatic disputes. By measuring the occurrence of such events over time, the GPR Index provides a quantitative indicator of geopolitical uncertainty and enables researchers to study its influence on macroeconomic variables and financial markets.

The GPR Index has been extensively used in academic research to analyse the relationship between geopolitical events and economic performance. It has demonstrated that periods of heightened geopolitical tension often correspond with increased market volatility, reduced investment activity, fluctuations in commodity prices, and changes in investor sentiment. Consequently, the GPR Index has become a benchmark for evaluating the economic impact of geopolitical developments.

Strengths

- Well-established and academically validated methodology.
- Quantifies geopolitical uncertainty using historical news data.
- Widely adopted in financial and macroeconomic research.
- Enables long-term trend analysis.

Limitations

- Primarily designed as a global index rather than an India-specific indicator.
- Relies heavily on keyword-based newspaper analysis.
- Does not provide real-time decision-support capabilities.

- Limited integration with modern AI-based Natural Language Processing techniques.

2.1.2.2 AI-GPR Index

Recent research, particularly the **AI-GPR Index** proposed by **Iacoviello and Tong** [2], extends traditional geopolitical risk measurement by incorporating Artificial Intelligence and Natural Language Processing techniques. Unlike conventional keyword-based approaches, the AI-GPR Index utilizes transformer-based language models and advanced text analysis techniques to extract contextual information from news articles.

These approaches improve the identification of complex geopolitical events and enable richer semantic understanding of news content. AI-driven geopolitical risk assessment is capable of analysing large volumes of textual information, identifying relationships among entities, and generating more adaptive geopolitical risk indicators compared to conventional keyword-based methods.

Strengths

- Captures contextual meaning of geopolitical events.
- Utilizes advanced NLP models for event understanding.
- More adaptable to evolving news content.
- Supports automated large-scale news analysis.

Limitations

- Computationally more expensive than traditional approaches.
- Performance depends on the quality of training data.
- Model predictions require explainability for user trust.
- Most implementations are research-oriented rather than end-user platforms.

2.1.2.3 Existing Geopolitical Monitoring Platforms

Several organizations have developed platforms that monitor geopolitical developments and provide strategic intelligence to governments, businesses, and investors. Platforms such as **Atlas Geopolitics** [14] and the **BBVA Geopolitics Monitor** [15] provide geopolitical reports, market intelligence, and economic analysis to support informed decision-making.

These platforms aggregate information from multiple data sources and present analytical reports describing emerging geopolitical trends and their potential economic

consequences. Although these systems provide valuable intelligence, they generally function as information services rather than integrated analytical platforms capable of combining geopolitical intelligence with predictive financial analytics.

Strengths

- Comprehensive global coverage.
- Professional analytical reporting.
- Continuous monitoring of geopolitical events.
- High-quality economic insights.

Limitations

- Limited transparency regarding analytical methodologies.
- Many services are subscription-based.
- Limited customization for individual users.
- Primarily descriptive rather than predictive.

2.1.2.4 AI-Based Financial Prediction Systems

Artificial Intelligence and Machine Learning techniques have been widely applied to predict financial market behaviour using historical market data, technical indicators, and external information sources. Machine learning algorithms such as **XGBoost** and other predictive models are capable of identifying complex relationships between market variables and improving forecasting performance across structured financial datasets.

While these systems demonstrate strong predictive capabilities under normal market conditions, many existing approaches rely predominantly on historical financial data and often fail to explicitly incorporate geopolitical developments into the prediction process. Consequently, unexpected geopolitical events may significantly reduce prediction reliability.

Strengths

- Supports predictive financial analytics.
- Learns complex market relationships.
- Improves forecasting accuracy through machine learning.
- Handles large financial datasets efficiently.

Limitations

- Heavy dependence on historical market behaviour.

- Limited integration of geopolitical intelligence.
- Reduced performance during unprecedented geopolitical crises.
- Prediction results may lack interpretability without explainable Artificial Intelligence techniques.

Table 2.2 Comparative Analysis of Existing Systems

System / Approach	Primary Objective	Major Strength	Major Limitation
Geopolitical Risk (GPR) Index	Quantify geopolitical uncertainty	Academically validated risk measurement	Limited India-specific analysis
AI-GPR Index	AI-driven geopolitical risk measurement	Improved contextual understanding of geopolitical events	High computational complexity
Commercial Geopolitical Monitoring Platforms	Monitor global geopolitical developments	Comprehensive geopolitical intelligence	Mostly descriptive and subscription-based
AI-Based Financial Prediction Systems	Predict market behaviour	High predictive capability	Limited integration of geopolitical intelligence

Discussion

Table 2.2 demonstrates that while existing systems provide effective solutions for individual aspects of geopolitical intelligence and financial analytics, none of the reviewed approaches offers a unified platform integrating all major analytical components.

The existing systems demonstrate significant progress in geopolitical risk measurement, news analysis, and financial prediction. However, each approach addresses only a specific aspect of the overall problem. Traditional geopolitical risk indices provide quantitative measures of uncertainty but lack contextual intelligence, whereas AI-based analytical systems improve event understanding without integrating comprehensive

financial decision-support functionalities. Commercial monitoring platforms provide valuable geopolitical information but generally focus on reporting rather than predictive analytics.

The literature therefore indicates the need for an integrated platform capable of combining geopolitical news intelligence, AI-driven event extraction, India-centric geopolitical risk assessment, financial market analysis, portfolio evaluation, supply chain risk assessment, and explainable analytical outputs. This requirement forms the motivation for the development of **Forsyt**.

2.1.3 Research Findings for Existing Literature

A comprehensive literature survey was conducted to study existing research related to geopolitical risk measurement, Artificial Intelligence-based geopolitical intelligence, and economic policy uncertainty. The reviewed literature helped establish the theoretical foundation of the proposed system and provided insights into the methodologies used for analysing geopolitical events and their influence on financial markets. The papers reviewed by the project team are summarized in **Table 2.3**.

Table 2.3 Research Findings from Existing Literature

S. No.	Roll Number	Name	Paper Title	Tools / Technology	Findings
1	102303629, 102303749	Aadi Jain, Devasya Kanwar	Iacoviello & Tong (2026) – <i>The AI-GPR Index: Measuring Geopolitical Risk using Artificial Intelligence</i>	Artificial Intelligence (AI), Natural Language Processing (NLP)	Proposed an AI-driven geopolitical risk index that improves contextual understanding of geopolitical events using Artificial Intelligence techniques.
2	102303909, 102303603	Vansh Gupta, Aaditi Verma	Caldara & Iacoviello (2022) – <i>Measuring Geopolitical Risk</i>	Newspaper-based Geopolitical Risk (GPR) Index	Developed a standardized methodology for measuring geopolitical risk using newspaper coverage and demonstrated that geopolitical risk significantly influences economic activity and financial markets.

3	102303934	Arianna Vohra	Baker, Bloom & Davis (2016) – <i>Measuring Economic Policy Uncertainty</i>	Newspaper-based Economic Policy Uncertainty (EPU) Index	Developed an index for measuring economic policy uncertainty and demonstrated its relationship with macroeconomic performance and financial market behaviour.
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Discussion

The literature survey indicates that geopolitical events and policy uncertainty have a significant impact on financial markets, investment behaviour, and overall economic stability. The work of **Caldara and Iacoviello** established a systematic framework for measuring geopolitical risk using newspaper-based analysis, while the **AI-GPR Index** demonstrated the application of Artificial Intelligence for enhancing geopolitical risk assessment through contextual analysis of news data. Similarly, the **Economic Policy Uncertainty (EPU) Index** highlighted the relationship between policy uncertainty and macroeconomic performance.

Although the reviewed studies provide robust methodologies for measuring geopolitical and economic uncertainty, they primarily focus on specific aspects of risk assessment. None of the reviewed literature presents an integrated, India-centric platform that combines geopolitical news intelligence, macroeconomic analysis, trade and supply chain intelligence, portfolio exposure assessment, and explainable Artificial Intelligence within a unified decision-support system. This identified research gap formed the basis for the development of **Forsyt**, which aims to integrate these analytical components into a single intelligent platform.

2.1.4 Problem Identified

The literature survey indicates that considerable progress has been made in developing intelligent memory architectures, collaborative AI systems, and long-term knowledge management frameworks. However, several challenges remain unresolved when these approaches are applied to geopolitical intelligence and financial decision-support systems.

The major problems identified from the literature survey are as follows:

- Existing memory architectures primarily focus on improving contextual reasoning and long-term knowledge retention rather than analysing geopolitical events and their impact on financial markets.
- Most existing systems are designed as **general-purpose AI frameworks** and are not specifically tailored for the Indian geopolitical and financial ecosystem.
- Current solutions do not provide an integrated platform capable of combining **news intelligence, geopolitical event extraction, risk quantification, market analysis, portfolio assessment, and supply chain risk analysis** within a single system.
- The majority of existing approaches emphasize conversational intelligence instead of structured decision-support for investors, analysts, and policymakers.
- Existing frameworks provide limited support for transforming continuous streams of geopolitical news into structured analytical insights that can be used for downstream financial analysis.
- Most surveyed systems focus on memory management or knowledge organization independently, with limited integration between Artificial Intelligence, Natural Language Processing, financial analytics, and visualization.
- The reviewed literature demonstrates limited emphasis on developing an **India-centric geopolitical risk assessment framework** capable of analysing the impact of global events on Indian financial markets.

These identified gaps collectively justify the development of **Forsyt**, which aims to integrate intelligent news analysis, structured geopolitical event extraction, India AI-GPR Index generation, predictive financial analytics, portfolio risk assessment, supply chain risk analysis, and interactive visualization into a unified geopolitical intelligence platform.

2.1.5 Survey of Tools and Technologies Used

The development of **Forsyt** requires the integration of multiple software tools, programming frameworks, machine learning libraries, databases, and deployment technologies. These technologies have been selected based on their ability to support geopolitical news analysis, Natural Language Processing (NLP), machine learning, data management, and interactive visualization. The major tools and technologies adopted during the development of the proposed system are discussed below.

2.1.5.1 Development Technologies

The core software development of Forsyt is carried out using Python for backend implementation and Git-based tools for collaborative software development.

Table 2.4 Development Technologies

Technology	Purpose in Forsyt
Python	Primary programming language used for backend development, data processing, Natural Language Processing, and Machine Learning implementation.
Git	Version control for tracking software changes and collaborative development.
GitHub	Repository hosting, source code management, project documentation, and team collaboration.
Visual Studio Code	Integrated Development Environment (IDE) used for software development, debugging, and testing.

2.1.5.2 Artificial Intelligence and Machine Learning Technologies

Artificial Intelligence forms the analytical foundation of Forsyt by enabling intelligent news processing, geopolitical risk estimation, predictive analytics, and explainable decision support.

Table 2.5 Artificial Intelligence and Machine Learning Technologies

Technology	Purpose in Forsyt
Hugging Face Transformers	Performs transformer-based Natural Language Processing for understanding and analysing geopolitical news.
XGBoost	Develops predictive models for analysing geopolitical influence on financial markets.

Scikit-learn	Supports data preprocessing, feature engineering, model training, and model evaluation.
PyTorch	Provides the deep learning framework required for transformer-based NLP models.
SHAP	Generates explainable AI outputs by identifying feature contributions towards model predictions.
Pandas	Data manipulation, preprocessing, and management of structured datasets.
NumPy	Numerical computation and efficient array processing during model development.

2.1.5.3 Data Storage and Communication Technologies

Forsyt requires efficient storage of processed information and seamless communication among software components.

Table 2.6 Data Storage and Communication Technologies

Technology	Purpose in Forsyt
PostgreSQL	Stores processed geopolitical events, analytical results, and application data.
REST APIs	Enable communication between the frontend interface and backend analytical modules.
JSON	Standard data exchange format used between different software components.

2.1.5.4 User Interface and Visualization Technologies

The analytical outputs generated by Forsyt are presented through an interactive dashboard that enables users to visualize geopolitical developments and financial insights.

Table 2.7 User Interface and Visualization Technologies

Technology	Purpose in Forsyt
React.js	Development of the interactive web-based dashboard and user interface.
Tailwind CSS	Responsive styling and layout of the web interface.
Recharts / D3.js	Interactive visualization of geopolitical trends, financial indicators, and analytical outputs.

2.1.5.5 Deployment and Supporting Technologies

To ensure maintainability, portability, and future scalability, Forsyt incorporates deployment and supporting technologies during software development.

Table 2.8 Deployment Technologies

Technology	Purpose in Forsyt
Docker	Containerized deployment to ensure consistent execution across different environments.
Docker Compose	Orchestrates the deployment of multiple application services, including PostgreSQL, the NLP pipeline, REST API, and React dashboard.

Google Cloud Run (optional)	Supports cloud deployment of Dockerized services for scalable production deployment.
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2.1.5.6 Data Acquisition Technologies

Technology	Purpose in Forsyt
feedparser	Retrieves geopolitical news from RSS feeds.
newspaper3k	Extracts textual content from news articles.
BeautifulSoup4	Parses HTML content during news extraction.
Scrapy	Supports large-scale web scraping and data collection.
yfinance	Retrieves historical financial market data for analysis.

Technology Selection Rationale

The technologies adopted for Forsyt were selected to provide a scalable, modular, and efficient software development environment capable of supporting geopolitical intelligence and financial analytics. Python was chosen because of its extensive ecosystem for Artificial Intelligence, Machine Learning, and Natural Language Processing. Hugging Face Transformers, XGBoost, Scikit-learn, and SHAP collectively enable intelligent news analysis, predictive modelling, and explainable AI, while Pandas and NumPy facilitate efficient data preprocessing and numerical computation.

PostgreSQL provides reliable storage for processed geopolitical information and analytical outputs, whereas REST APIs enable communication between the backend services and the interactive frontend developed using React.js. React.js, together with Tailwind CSS and Recharts/D3.js, provides an interactive and responsive interface for visualizing geopolitical trends, financial indicators, and analytical outputs. Git and

GitHub facilitate collaborative software development, while Docker provides a consistent deployment environment across different platforms.

Overall, the selected technologies form an integrated software ecosystem that supports the complete analytical workflow of Forsyt, from geopolitical news acquisition and processing to predictive analytics, explainable AI, and interactive visualization.

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2.2 Software Requirement Specification

2.2.1 Introduction

2.2.1.1 Purpose

The purpose of this Software Requirement Specification (SRS) is to define the software requirements for Forsyt, an AI-powered geopolitical intelligence platform developed to analyse the impact of global geopolitical events on the Indian financial ecosystem. This document provides a comprehensive description of the system's objectives, scope, software features, interfaces, and non-functional requirements, thereby serving as a reference throughout the development lifecycle.

The SRS establishes a common understanding of the proposed system among all stakeholders involved in the project, including the development team, project mentors, evaluators, and future contributors. It serves as a guideline during system design, implementation, testing, validation, and maintenance by ensuring that all software components are developed according to predefined requirements.

Furthermore, this document defines the expected behaviour of the system and provides a structured framework for verifying whether the implemented software satisfies the functional objectives of the Forsyt platform. By documenting the software requirements before implementation, the project aims to improve consistency, maintainability, and overall software quality.

2.2.1.2 Intended Audience and Reading Suggestions

This document is intended for multiple stakeholders associated with the development and evaluation of Forsyt. Each stakeholder may refer to different sections of the report depending on their role and area of interest.

Table 2.8 Intended Audience

Stakeholder	Purpose of Reading
Development Team	To understand the software requirements, architecture, and implementation details required during development.

Project Mentors	To monitor project progress, evaluate software design decisions, and verify compliance with project objectives.
Evaluation Panel	To assess whether the developed system satisfies the specified requirements and project scope.
Researchers and Students	To understand the design methodology and implementation approach adopted for the proposed system.
Future Developers	To facilitate future maintenance, feature enhancement, and further development of the software.

Readers are encouraged to first review **Chapter 1 (Introduction)** to understand the motivation and objectives of the project before studying the Software Requirement Specification. The remaining sections should be read sequentially, as each section builds upon the concepts introduced in the previous one.

2.2.1.3 Project Scope

Forsyt is designed as an **India-centric Geopolitical Risk Intelligence Platform** that transforms unstructured geopolitical news into structured analytical insights using Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML). The platform aims to assist users in understanding how geopolitical developments influence Indian financial markets by providing risk analysis, predictive analytics, and interactive visualizations.

The software continuously processes geopolitical news, extracts significant events through an NLP pipeline, and generates the **India AI-GPR Index**, which serves as a quantitative indicator of geopolitical risk. The generated insights are further utilized by multiple analytical modules, including Nifty 50 market analysis, Portfolio Risk Advisor, and Supply Chain Risk Assessment, to support informed decision-making.

The scope of the current implementation includes the following major functionalities:

- Collection and processing of geopolitical news from multiple sources.
- NLP-based extraction of structured geopolitical events.
- Generation of the India AI-GPR Index.
- Analysis of the relationship between geopolitical developments and Nifty 50 market behaviour.
- Portfolio Risk Assessment based on geopolitical exposure.
- Supply Chain Risk Assessment for major trade corridors.
- Interactive dashboard for visualization of analytical outputs.

- Explainable AI techniques to improve transparency and interpretation of analytical results.

Forsyt is intended to function as a **decision-support system** that assists users in analysing geopolitical developments and their financial implications. It does not execute financial transactions, provide automated investment recommendations, or replace human decision-making.

2.2.2 Overall Description

2.2.2.1 Product Perspective

Forsyt is an AI-powered geopolitical intelligence platform developed to assist users in understanding the impact of global geopolitical events on the Indian financial ecosystem. The system is designed as a decision-support application that combines Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML) to transform unstructured geopolitical news into structured analytical insights. Rather than functioning as a conventional news aggregation platform, Forsyt analyses geopolitical information to identify meaningful events, quantify geopolitical risk, and present actionable insights through an interactive dashboard.

The software follows a modular architecture in which multiple analytical components work together to perform specialized tasks. The workflow begins with the collection of geopolitical news from multiple sources. The acquired news is processed through an NLP pipeline to extract relevant entities, relationships, and events. The processed information is then utilized to generate the India AI-GPR (Artificial Intelligence Geopolitical Risk) Index, which forms the foundation for subsequent analytical modules, including Nifty 50 market analysis, Portfolio Risk Advisor, and Supply Chain Risk Assessment.

Each module is designed to operate independently while exchanging information through well-defined software interfaces. This modular approach improves maintainability, scalability, and future extensibility by allowing individual components to be updated or enhanced without significantly affecting the remaining system. The architecture also supports the integration of additional analytical modules as future project extensions.

The processed analytical outputs are presented through an interactive dashboard that enables users to visualize geopolitical events, monitor risk indicators, analyse financial

implications, and interpret model predictions using explainable AI techniques. The platform is intended to support informed decision-making by providing organized and data-driven analytical insights. However, it does not replace human judgment or perform autonomous investment decisions.

The overall architecture and interaction among the major software components are illustrated in **Figure 2.1**.

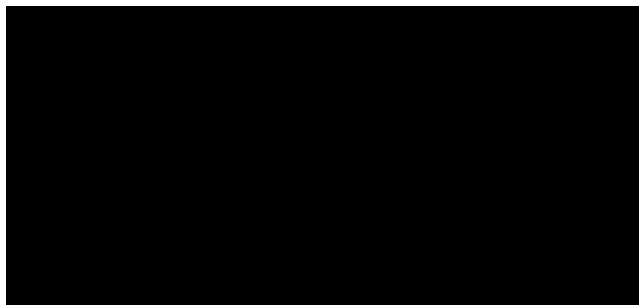


Figure 2.1 Overall Product Perspective of Forsyt

(Insert Overall System Architecture Diagram here.)

Figure 2.1 illustrates the overall software architecture of Forsyt. The system begins with geopolitical news collection, followed by NLP-based event extraction and structured data generation. The extracted information is used to compute the India AI-GPR Index, which supports multiple analytical modules, including market analysis, portfolio risk assessment, and supply chain risk assessment. The outputs from these modules are integrated and presented through the dashboard, enabling users to interpret geopolitical developments and their potential impact on the Indian financial ecosystem.

2.2.2.2 Product Features

Forsyt is designed as an integrated geopolitical intelligence and financial decision-support platform comprising multiple analytical modules. Each module performs a specific function while interacting with other components to provide a comprehensive analysis of geopolitical developments and their potential impact on the Indian financial ecosystem. The major product features of the proposed system are summarized in **Table 2.9**

Table 2.9 Major Product Features of Forsyt

Product Feature	Description
News Intelligence System	Collects and processes geopolitical news from selected sources for further analysis.
NLP-Based Event Extraction	Identifies geopolitical entities, events, relationships, and contextual information from unstructured news articles.
India AI-GPR Index Generation	Generates an India-centric Artificial Intelligence Geopolitical Risk Index based on extracted geopolitical events.
Nifty 50 Market Analysis	Analyses the relationship between geopolitical developments and Nifty 50 market behaviour to support financial analysis.
Portfolio Risk Advisor	Evaluates the geopolitical exposure of investment portfolios by analysing sector-wise risks.
Supply Chain Risk Assessment	Assesses the impact of geopolitical events on major trade routes and supply chain networks relevant to India.
Explainable AI Module	Provides interpretable explanations for analytical outputs to improve transparency and user understanding.
Interactive Dashboard	Presents geopolitical events, India AI-GPR trends, market insights, and risk assessments through graphical visualizations and reports.
REST API Integration	Enables communication between the backend analytical modules and the frontend dashboard through standardized APIs.

The above features collectively enable Forsyt to transform large volumes of unstructured geopolitical information into structured analytical intelligence. Rather than functioning as independent components, these modules operate as an integrated software platform where the output of one module serves as the input for subsequent analytical processes. This modular design improves maintainability, scalability, and future extensibility while ensuring that each component performs a clearly defined function within the overall system.

2.2.3 External Interface Requirements

2.2.3.1 User Interfaces

The user interface of Forsyt is designed to provide users with an intuitive and interactive environment for analysing geopolitical developments and their impact on the Indian financial ecosystem. The interface enables users to access different analytical modules through a centralized dashboard, allowing seamless navigation between various functionalities without requiring technical expertise.

The interface presents information using graphical visualizations, charts, tables, and interactive components to improve the interpretation of analytical results. Users can monitor geopolitical events, observe trends in the India AI-GPR Index, analyse market behaviour, evaluate portfolio exposure, and assess supply chain risks through dedicated dashboard modules.

The major user interface components of Forsyt are summarized in **Table 2.10**.

Table 2.10 User Interface Components

Component	Description
Dashboard Home	Displays an overview of the platform along with key geopolitical and financial indicators.
News Intelligence Module	Presents processed geopolitical news and extracted event information.
India AI-GPR Dashboard	Displays the generated India AI-GPR Index along with trend visualizations.
Nifty 50 Analysis Module	Visualizes market analysis and volatility insights derived from geopolitical events.
Portfolio Risk Advisor	Displays portfolio risk assessments based on geopolitical exposure.
Supply Chain Risk Module	Presents trade route and supply chain risk analysis.
Explainable AI View	Displays explanations supporting generated analytical outputs and predictions.

The user interface emphasizes simplicity, responsiveness, and ease of navigation to ensure that users can efficiently interpret analytical information and access different software modules. The interface is designed to support future expansion by allowing

additional analytical modules to be incorporated without significant modification to the existing layout.

2.2.3.2 Hardware Interfaces

Forsyt is a software-based application and does not require any specialized hardware interfaces or dedicated embedded devices for its operation. The platform is designed to execute on standard computing systems capable of supporting Artificial Intelligence, Natural Language Processing, and Machine Learning workloads. All interactions with the system are performed through conventional input and output devices connected to the host computer.

The minimum hardware requirements for the proposed system are summarized in **Table 2.11**.

Table 2.11 Hardware Requirements

Hardware Component	Specification
Processor	Intel Core i5 (10th Generation or above) / AMD Ryzen 5 or equivalent
RAM	Minimum 8 GB (16 GB recommended)
Storage	Minimum 20 GB available disk space
Graphics	Integrated Graphics (Dedicated GPU recommended for model training)
Display	1366 × 768 resolution or higher
Input Devices	Keyboard and Mouse
Internet Connectivity	Required for news acquisition, software updates, and external data access

The system communicates with users through standard display devices and receives user inputs via a keyboard and mouse. Since Forsyt primarily performs software-based analytical tasks, no custom hardware interfaces or peripheral communication devices are required. However, higher hardware configurations may improve processing performance during machine learning model training and large-scale data analysis.

2.2.3.3 Software Interfaces

Forsyt integrates multiple software frameworks, libraries, databases, and external services to support news collection, Natural Language Processing (NLP), machine learning, data storage, and visualization. These software interfaces enable seamless communication between different components of the platform and ensure efficient execution of the complete analytical pipeline. The major software interfaces used in the proposed system are summarized in **Table 2.12**.

Table 2.12 Software Interfaces

Software Interface	Purpose
Python	Primary programming language used for backend development, NLP processing, and machine learning implementation.
React.js	Develops the interactive web-based dashboard for visualization and user interaction.
REST APIs	Facilitate communication between backend analytical modules and the frontend dashboard.
PostgreSQL	Stores processed geopolitical events, analytical outputs, user-related information (if applicable), and system data.
Hugging Face Transformers	Provides pre-trained transformer models used for NLP-based event extraction and text understanding.
XGBoost	Provides predictive models used by the backend to estimate market volatility and geopolitical risk.
Scikit-learn	Supports data preprocessing, feature engineering, model evaluation, and auxiliary machine learning tasks.
PyTorch	Supports deep learning model development and execution where required.
SHAP	Interfaces with the prediction module to generate feature-level explanations for model outputs.
News Collection Module (feedparser / newspaper3k / BeautifulSoup4 / Scrapy)	Collect and process geopolitical news from RSS feeds and web sources.
yfinance	Retrieves historical financial market data for analytical modules.

Docker	Provides containerized deployment for consistent execution across different computing environments.
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The software components operate together to form an integrated analytical platform. The backend processes geopolitical news, generates structured intelligence, performs predictive analysis, and exposes the generated results through REST APIs. The frontend dashboard retrieves these results and presents them through interactive visualizations, while PostgreSQL maintains persistent storage of processed information and analytical outputs. The overall software ecosystem ensures modularity, interoperability, and scalability, enabling independent development and future enhancement of individual system components. The technologies listed above are consistent with the project's documented technology stack and deployment architecture.

2.2.4 Other Non-functional Requirements

2.2.4.1 Performance Requirements

The performance requirements define the expected operational characteristics of Forsyt to ensure efficient processing of geopolitical data and timely delivery of analytical insights. The system should maintain acceptable response times while handling data ingestion, Natural Language Processing, machine learning inference, and dashboard visualization.

The expected performance requirements are summarized in **Table 2.13**.

Table 2.13 Performance Requirements

Requirement	Expected Performance
Dashboard Response Time	Less than 500 ms for dashboard requests under normal operating conditions.
News Processing	Process newly collected news articles through the NLP pipeline in a timely manner according to the configured ingestion schedule.
API Response Time	REST API endpoints should respond within acceptable latency under normal workload.

Database Performance	Efficient retrieval and storage of geopolitical events and analytical outputs.
Scalability	Support future integration of additional analytical modules and news sources without major architectural changes.
Availability	The system should remain operational throughout scheduled analysis and dashboard usage.

The platform is designed using a modular architecture that enables independent execution of software components while maintaining overall system efficiency. Performance optimization techniques are incorporated during development to improve analytical processing and dashboard responsiveness. The project roadmap also targets an API response time of **less than 500 ms** during deployment optimization.

2.2.4.2 Safety Requirements

Since Forsyt is an information and decision-support platform, it does not directly control physical devices or perform safety-critical operations. However, the following safety requirements have been considered during system development to ensure reliable operation and responsible presentation of analytical results.

- The system shall present generated analytical outputs as **decision-support information** rather than definitive financial advice.
- The platform shall notify users that analytical predictions are dependent on the quality and availability of input data.
- The system shall continue operating even if an individual news source becomes temporarily unavailable by utilizing available data sources or configured fallback mechanisms.
- Input validation shall be performed to reduce the possibility of processing invalid or incomplete data.
- System errors and processing failures shall be logged to facilitate debugging and maintenance.
- The platform shall avoid modification or deletion of stored analytical data without appropriate administrative control.

These requirements contribute to the reliable and stable operation of the software while minimizing the impact of unexpected processing failures or incomplete datasets.

2.2.4.3 Security Requirements

Forsyt incorporates appropriate software security practices to protect system resources, maintain data integrity, and support secure communication between software components.

The principal security requirements are listed below:

- Communication between frontend and backend modules shall occur through standardized REST APIs.
- Database access shall be restricted to authorized backend services.
- Configuration parameters and sensitive credentials shall be maintained using environment configuration files instead of hard-coded values.
- User inputs shall be validated before processing to reduce the possibility of invalid or malicious requests.
- System logs shall be maintained to assist in monitoring software execution and identifying operational issues.
- The application shall maintain the integrity of stored geopolitical events and analytical results throughout processing.

Since Forsyt is developed as an academic decision-support platform, it does not currently process highly sensitive personal information or financial transactions. Nevertheless, secure software development practices have been adopted to support future scalability and deployment in more demanding operational environments.

2.3 Cost Analysis

The development of Forsyt primarily involves software development, data processing, machine learning model implementation, and deployment activities. Since the project is developed as an academic capstone project, several software tools and frameworks used during development are open-source and therefore do not incur licensing costs. The estimated project cost mainly consists of hardware resources, cloud deployment, internet usage, and miscellaneous development expenses.

2.3.1 Software Cost

The software components used in Forsyt are largely based on open-source technologies, significantly reducing the overall development expenditure.

Table 2.14 Software Cost Estimation

Software/Tool	License Type	Estimated Cost (INR)
Python	Open Source	0
React.js	Open Source	0
PostgreSQL	Open Source	0
PyTorch	Open Source	0
Scikit-learn	Open Source	0
Hugging Face Transformers	Open Source	0
SHAP	Open Source	0
Git & GitHub	Free (Student/Educational)	0
Docker	Community Edition	0

Total Software Cost: ₹0

2.3.2 Hardware Cost

The project is developed using standard computing devices without requiring specialized hardware.

Table 2.15 Hardware Cost Estimation

Resource	Estimated Cost (INR)
Personal Laptop/Desktop	Existing Resource
Internet Connectivity	Existing Resource
External Storage)	1,500
Miscellaneous Accessories	1,000

Estimated Hardware Cost: ₹2,500

2.3.3 Cloud and Deployment Cost

Forsyt supports deployment using cloud platforms. During academic development, free-tier services are utilized wherever possible.

Table 2.16 Cloud Deployment Cost

Service	Estimated Cost (INR)
Google Cloud Run (Free Tier)	0
Docker Deployment	0
GitHub Repository Hosting	0
Domain Registration	1,000/year

Estimated Cloud Cost: ₹1,000 (Optional)

2.3.4 Human Resource Cost

Since Forsyt is developed as part of the B.E. Capstone Project by the student development team, no direct labour cost is incurred. Faculty mentorship is provided as part of the academic curriculum and is therefore excluded from the financial estimation.

2.3.5 Overall Cost Estimation

Table 2.17 Overall Project Cost

Category	Estimated Cost (INR)
Software	0
Hardware	2,500
Cloud Services	1,000
Human Resource	Academic Project
Miscellaneous	500

Estimated Total Project Cost: ₹4,000

Cost Analysis Summary

The estimated development cost of Forsyt is relatively low because the project extensively utilizes open-source software frameworks, freely available development tools, and existing computing infrastructure. The primary expenditure is associated with optional cloud deployment and minor hardware-related expenses rather than software licensing. This makes the proposed platform economically feasible for academic development while also providing flexibility for future deployment and scalability.

2.4 Risk Analysis

Risk analysis is an essential activity in software engineering that involves identifying potential risks that may affect the successful development, deployment, and operation of a software system. Early identification of these risks enables the development team to implement appropriate mitigation strategies and minimize their impact on project objectives.

For the Forsyt platform, risks have been analysed across technical, operational, data, and project management aspects. The identified risks and their corresponding mitigation strategies are presented in **Table 2.18**.

Table 2.18 Risk Assessment Matrix

Risk Description	Probability	Impact	Mitigation Strategy
Delay in project development due to academic schedule and time constraints.	Medium	High	Follow a structured development plan with regular milestone reviews.
Poor quality or incomplete geopolitical news data affecting analytical accuracy.	Medium	High	Use reliable news sources and perform data preprocessing and validation.
Machine learning models may not achieve the expected prediction accuracy.	Medium	Medium	Perform feature engineering, model evaluation, and hyperparameter tuning.

Changes in APIs or external data sources may interrupt data collection.	Low	High	Design the system with modular data collection components and support alternative sources where possible.
Increased processing time due to large volumes of geopolitical news.	Medium	Medium	Optimize preprocessing, indexing, and model inference pipelines.
Integration issues between frontend and backend components.	Low	Medium	Conduct regular integration testing and API validation throughout development.
Database failure or data corruption.	Low	High	Maintain periodic backups and implement proper database management practices.
Unauthorized access to project configuration or sensitive credentials.	Low	High	Store credentials securely using environment variables and restrict repository access.
Software defects affecting dashboard functionality.	Medium	Medium	Perform systematic testing and debugging before deployment.
Future scalability challenges as analytical modules expand.	Medium	Medium	Maintain a modular architecture that supports independent component enhancement.

Risk Classification

The identified risks can be categorized as follows:

Technical Risks

- Model performance may not meet expected analytical accuracy.
- High computational requirements for NLP and machine learning models.
- Software integration challenges among different analytical modules.

Data Risks

- Incomplete, delayed, or inconsistent geopolitical news.

- Dependency on external datasets and APIs.
- Data preprocessing errors affecting downstream analysis.

Project Risks

- Limited development time due to academic schedules.
- Coordination challenges among team members.
- Delays in testing, documentation, or deployment activities.

Operational Risks

- Temporary unavailability of external services.
- Hardware limitations during model training.
- Network connectivity affecting data collection and dashboard access.

Risk Mitigation Strategy

The modular architecture adopted in Forsyt significantly reduces technical and maintenance risks by allowing individual software components to be developed, tested, and updated independently. Regular code reviews, version control using GitHub, incremental testing, and continuous validation of analytical modules further improve software reliability throughout the development lifecycle.

To minimize data-related risks, the system incorporates preprocessing and validation mechanisms before analytical processing. Software security risks are addressed through controlled access to configuration files and proper management of sensitive credentials. The use of open-source technologies, modular APIs, and scalable software design also reduces future maintenance and deployment risks.

Overall, the identified risks are considered manageable within the scope of the project and can be effectively mitigated through appropriate software engineering practices and systematic project management.

3. METHODOLOGY ADOPTED

3.1 Investigative Techniques

The development of Forsyt followed a systematic investigative approach to understand how geopolitical events influence the Indian financial ecosystem and how Artificial

Intelligence can be used to transform unstructured geopolitical news into actionable financial intelligence. Instead of directly implementing predictive models, the project first investigated existing geopolitical risk measurement techniques, analysed available datasets, evaluated suitable AI models, and validated the proposed methodology against established research.

The investigative process adopted during the development of Forsyt is illustrated in **Figure 3.1**.



Figure 3.1 Investigative Methodology

(Insert investigative process flowchart here.)

Phases of Investigation

Phase 1: Literature Investigation

The project began with an extensive review of existing research to understand:

- Geopolitical Risk (GPR) Index proposed by Caldara & Iacoviello.
- AI-based Geopolitical Risk (AI-GPR) approaches.
- Economic Policy Uncertainty (EPU).
- Financial market prediction techniques.
- Explainable AI methods.

Outcome: Established the theoretical foundation and identified research gaps.

Phase 2: Problem Investigation

Existing geopolitical monitoring systems and financial prediction models were analysed to identify their limitations.

The investigation revealed that:

- Existing GPR indices are largely global in scope.
- Most systems focus either on news monitoring or financial prediction, but not both.
- Few solutions integrate explainable AI.
- Limited emphasis is placed on the Indian financial ecosystem.

Outcome: Defined the objectives of Forsyt.

Phase 3: Dataset Investigation

Relevant datasets were studied to determine their suitability for geopolitical risk analysis.

The investigation focused on:

- Geopolitical news sources.
- Historical financial market data.
- Data quality.
- Data availability.
- Temporal consistency.

Outcome: Selected datasets for model development and validation.

Phase 4: Model Investigation

Different analytical approaches were evaluated before selecting the final methodology.

The investigation included:

- NLP-based event extraction.
- Transformer-based language models.
- India AI-GPR Index construction.
- Machine learning models for market analysis.
- Explainable AI techniques.

Outcome: Selected the analytical pipeline adopted in Forsyt.

Phase 5: Validation Investigation

The proposed system was evaluated using multiple validation techniques.

These included:

- Comparison with established geopolitical risk methodologies.
- Performance evaluation of NLP models.
- Prediction analysis.
- Explainability assessment.

Outcome: Verified the reliability of the proposed platform.

Summary of Investigative Process

Table 3.1 Investigative Process

Phase	Objective	Outcome
Literature Investigation	Study existing research	Identify theoretical foundation
Problem Investigation	Analyse current systems	Identify research gap
Dataset Investigation	Evaluate available datasets	Select appropriate data
Model Investigation	Compare AI approaches	Design analytical pipeline
Validation Investigation	Evaluate developed system	Verify effectiveness

Justification for the Selected Investigative Technique

The investigative approach adopted for Forsyt combines **research-oriented analysis** with **iterative software development**. This methodology was selected because the project required understanding existing geopolitical risk methodologies before designing an India-centric analytical platform. The phased investigation ensured that every major design decision—from dataset selection to AI model development and validation—was supported by literature, experimental evaluation, and implementation requirements. This systematic approach improved the reliability, scalability, and interpretability of the final system.

3.2 Proposed Solution

The proposed solution, **Forsyt**, is designed as an AI-powered geopolitical intelligence platform that transforms unstructured geopolitical news into structured analytical

insights for the Indian financial ecosystem. The system integrates Natural Language Processing (NLP), Machine Learning (ML), explainable AI, and interactive visualization to analyse geopolitical developments and evaluate their potential impact on financial markets. Unlike conventional geopolitical monitoring platforms that primarily provide descriptive reports, Forsyt combines automated news intelligence, geopolitical risk estimation, financial analytics, and decision-support capabilities within a unified software platform.

The proposed solution follows a modular architecture in which each functional component performs a specialized task while contributing to the overall analytical workflow. Information generated by one module serves as the input for subsequent modules, enabling the system to progressively transform raw geopolitical news into meaningful financial intelligence.

The overall workflow of the proposed solution is illustrated in **Figure 3.2**.

Figure 3.2 Proposed Solution Workflow

(Insert the overall system workflow diagram here.)



Workflow of the Proposed Solution

The proposed solution is implemented through the following sequential stages:

Stage 1: Geopolitical News Acquisition

- Collect geopolitical news from multiple news sources.
- Aggregate articles related to international politics, conflicts, diplomacy, trade, and economic events.
- Store collected articles for further processing.

Output: Raw geopolitical news articles.

Stage 2: News Preprocessing and NLP

- Clean and preprocess collected news articles.
- Remove irrelevant textual information.
- Apply transformer-based Natural Language Processing models.
- Extract meaningful geopolitical entities and events.

Output: Structured geopolitical event data.

Stage 3: India AI-GPR Index Generation

- Analyse extracted geopolitical events.
- Estimate the intensity of geopolitical uncertainty.
- Generate the India AI-GPR Index representing daily geopolitical risk.

Output: India AI-GPR Index.

Stage 4: Financial Analytics

The generated India AI-GPR Index is utilized by multiple analytical modules:

- **Nifty 50 Analysis** – evaluates the relationship between geopolitical developments and market behaviour.
- **Portfolio Risk Advisor** – estimates geopolitical exposure across investment portfolios.
- **Supply Chain Risk Assessment** – analyses geopolitical risks affecting major trade routes and logistics.

Output: Analytical insights for multiple financial applications.

Stage 5: Explainable AI

- Generate explanations supporting machine learning predictions.
- Identify influential features contributing to analytical outputs.
- Improve transparency and interpretability of generated insights.

Output: Explainable analytical results.

Stage 6: Interactive Dashboard

- Present geopolitical events.
- Display India AI-GPR trends.
- Visualize market analysis and risk indicators.
- Provide users with an integrated decision-support interface.

Output: Interactive visualization of analytical results.

Major Functional Modules

Table 3.2 Functional Modules of Forsyt

Module	Primary Function	Output
News Intelligence Module	Collects and preprocesses geopolitical news	Raw news dataset
NLP Event Extraction Module	Extracts structured geopolitical events	Structured event data
India AI-GPR Module	Computes geopolitical risk indicator	India AI-GPR Index
Nifty 50 Analysis Module	Analyses market behaviour	Market insights
Portfolio Risk Advisor	Evaluates geopolitical exposure	Portfolio risk assessment
Supply Chain Risk Module	Assesses logistics and trade risks	Supply chain risk analysis
Explainable AI Module	Explains analytical predictions	Interpretable results
Interactive Dashboard	Presents analytical outputs	User-friendly visualization

Advantages of the Proposed Solution

The proposed solution offers the following advantages over existing approaches:

- Integrates multiple analytical modules within a single platform.
- Combines geopolitical intelligence with financial market analysis.

- Generates an India-centric geopolitical risk indicator.
- Utilizes AI-based Natural Language Processing for event understanding.
- Supports explainable AI to improve transparency and user confidence.
- Provides an interactive dashboard for real-time visualization and analysis.
- Employs a modular architecture that supports future scalability and extension.

Summary

The proposed solution integrates geopolitical news intelligence, Natural Language Processing, India AI-GPR Index generation, financial analytics, explainable AI, and interactive visualization into a unified decision-support platform. The modular design enables efficient interaction among individual components while ensuring scalability, maintainability, and ease of future enhancement. By transforming unstructured geopolitical information into structured analytical intelligence, Forsyt provides users with a comprehensive platform for understanding the potential financial implications of geopolitical developments.

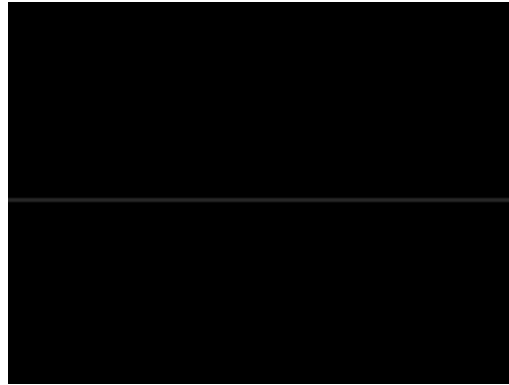
3.3 Work Breakdown Structure

The development of Forsyt was organized into a structured Work Breakdown Structure (WBS) to divide the project into manageable work packages. This approach enabled systematic planning, task allocation, progress monitoring, and integration of individual software components. Each work package represents a major functional area of the project and contributes to the successful implementation of the complete geopolitical intelligence platform.

The overall Work Breakdown Structure of Forsyt is illustrated in **Figure 3.3**.

Figure 3.3 Work Breakdown Structure of Forsyt

(Insert WBS Diagram here.)



Major Work Packages

The project was divided into the following major work packages.

WP1 – Literature Review and Requirement Analysis

Activities

- Study geopolitical risk literature.
- Analyse Caldara's GPR methodology.
- Review AI-GPR approaches.
- Identify system requirements.
- Define project objectives.

Deliverables

- Literature Survey
- Requirement Analysis
- Software Requirement Specification (SRS)

WP2 – Data Collection and Preprocessing

Activities

- Collect geopolitical news.
- Acquire financial market data.
- Clean and preprocess textual data.
- Prepare datasets for analysis.

Deliverables

- Processed news dataset
- Structured financial dataset

WP3 – NLP and Event Extraction Module

Activities

- Perform text preprocessing.
- Extract geopolitical entities.
- Identify significant events.
- Generate structured event information.

Deliverables

- Structured geopolitical events
- NLP processing pipeline

WP4 – India AI-GPR Index Development

Activities

- Analyse extracted geopolitical events.
- Compute geopolitical risk scores.
- Generate India AI-GPR Index.
-

Deliverables

- India AI-GPR Index
- Risk trend analysis

WP5 – Financial Analytics Module

Activities

- Analyse Nifty 50 behaviour.
- Evaluate portfolio exposure.
- Perform supply chain risk analysis.

- Generate analytical insights.

Deliverables

- Market analysis
- Portfolio Risk Advisor
- Supply Chain Risk Assessment

WP6 – Dashboard and System Integration

Activities

- Develop interactive dashboard.
- Integrate backend modules.
- Connect APIs.
- Display analytical outputs.

Deliverables

- Interactive Dashboard
- Integrated software platform

WP7 – Testing and Validation

Activities

- Validate analytical modules.
- Evaluate prediction performance.
- Perform integration testing.
- Debug identified issues.

Deliverables

- Tested software
- Validation results

WP8 – Documentation and Final Deployment

Activities

- Prepare technical documentation.
- Finalize project report.
- Deploy application.
- Demonstrate completed system.

Deliverables

- Technical Report
- Final Software Prototype

Summary of Work Packages

Table 3.3 Work Breakdown Summary

Work Package	Description	Major Deliverable
WP1	Literature Review & Requirement Analysis	SRS and Literature Survey
WP2	Data Collection & Preprocessing	Processed datasets
WP3	NLP Event Extraction	Structured geopolitical events
WP4	India AI-GPR Development	India AI-GPR Index
WP5	Financial Analytics	Risk analysis modules
WP6	Dashboard & Integration	Interactive Dashboard
WP7	Testing & Validation	Verified system
WP8	Documentation & Deployment	Final software and report

Discussion

The Work Breakdown Structure ensured that the project was developed incrementally, with each work package producing clearly defined deliverables that supported the subsequent phase of development. The modular organization also enabled parallel development of analytical components, simplified testing and integration, and improved overall project management. By dividing the project into independent yet interconnected work packages, the development team was able to systematically progress from

literature review and data preparation to implementation, validation, and deployment of the complete Forsyt platform.

3.4 Tools and Technology

The successful implementation of Forsyt required the integration of multiple software technologies for application development, data processing, Natural Language Processing (NLP), machine learning, database management, and visualization. The selected technologies were chosen based on their compatibility, scalability, and ability to support the analytical requirements of an AI-powered geopolitical intelligence platform. Each technology contributed to a specific stage of the system's implementation, from data acquisition to interactive visualization.

Technology Stack Used in Forsyt

Table 3.4 Technology Stack

Technology	Role in Implementation
Python	Core programming language used for implementing backend services, data processing, machine learning, and NLP workflows.
React.js	Developed the interactive web interface used for displaying geopolitical insights and analytical outputs.
PostgreSQL	Stored processed geopolitical events, analytical results, and application data.
REST APIs	Enabled communication between frontend and backend components for data exchange.
Hugging Face Transformers	Implemented transformer-based Natural Language Processing for geopolitical news analysis and event extraction.
XGBoost	Developed predictive models for analysing the relationship between geopolitical developments and financial markets.
Scikit-learn	Supported preprocessing, feature engineering, model evaluation, and auxiliary machine learning tasks.
PyTorch	Provided the deep learning framework required for transformer-based NLP models.
SHAP	Generated explainable AI outputs to improve transparency and interpretability of predictive models.

Recharts / D3.js	Created interactive visualizations for geopolitical trends, the India AI-GPR Index, portfolio analytics, and financial market indicators.
Tailwind CSS	Developed a responsive and visually consistent user interface for the dashboard.
Git & GitHub	Managed source code, version control, and collaborative software development.

Technology Usage During Development

The implementation of Forsyt was divided into multiple functional layers, with each technology supporting a specific aspect of the software architecture.

Backend Development

- Python served as the primary backend programming language.
- REST APIs enabled communication between software modules.
- PostgreSQL managed persistent storage of processed data.

Artificial Intelligence and NLP

- Hugging Face Transformers processed geopolitical news articles.
- PyTorch supported transformer-based model execution.
- Scikit-learn handled preprocessing and model evaluation.
- XGBoost implemented predictive analytical models.
- SHAP generated explanations for machine learning predictions.

Frontend and Visualization

- React.js implemented the web-based user interface.
- Recharts / D3.js provided interactive charts and visualizations for geopolitical trends, the India AI-GPR Index, and financial market analytics.
- Tailwind CSS enabled responsive styling and a consistent user interface across dashboard components.

Software Development and Collaboration

- Git tracked software revisions throughout development.
- GitHub facilitated collaborative coding, repository management, and documentation.

Technology Integration

The technologies were integrated to create a modular software architecture. News data collected by the backend is processed using Natural Language Processing models before being analysed by machine learning components. The processed results are stored in PostgreSQL and exposed through REST APIs. The frontend dashboard retrieves these results and presents them using interactive visualizations, enabling users to analyse geopolitical developments and their potential financial implications through a unified interface.

Advantages of the Selected Technology Stack

The adopted technology stack provides several advantages:

- Modular and scalable software architecture.
- Efficient processing of large volumes of geopolitical news.
- Reliable storage and retrieval of analytical data.
- Interactive visualization of geopolitical and financial insights.
- Explainable AI for improved transparency.
- Ease of maintenance through version control and collaborative development.

4. DESIGN SPECIFICATIONS

4.1 System Architecture

The architecture of Forsyt is designed as a modular, scalable, and AI-driven analytical framework that converts unstructured geopolitical news into structured financial intelligence for the Indian market. Rather than functioning as a conventional news aggregation platform, the system integrates automated data acquisition, Natural Language Processing (NLP), geopolitical risk modelling, machine learning, explainable AI, and interactive visualization within a unified software ecosystem. This modular architecture allows individual components to perform specialized analytical tasks while collectively supporting the end-to-end generation of geopolitical intelligence. The repository also documents this layered implementation approach through separate ingestion, NLP, machine learning, API, dashboard, and deployment components.

The architecture follows a layered processing pipeline in which raw geopolitical news is progressively transformed into structured, interpretable, and actionable information. Information collected from news sources undergoes preprocessing and NLP-based event extraction before being used for India AI-GPR Index generation. The generated geopolitical risk information subsequently supports multiple downstream analytical modules, including Nifty 50 volatility analysis, Portfolio Risk Advisor, and Supply Chain Risk Assessment. The processed analytical outputs are finally delivered to users through an interactive dashboard supported by REST APIs and a PostgreSQL database.

Overall System Architecture

The overall architecture of Forsyt is illustrated in **Figure 4.1**.

I recommend creating the diagram exactly like this:

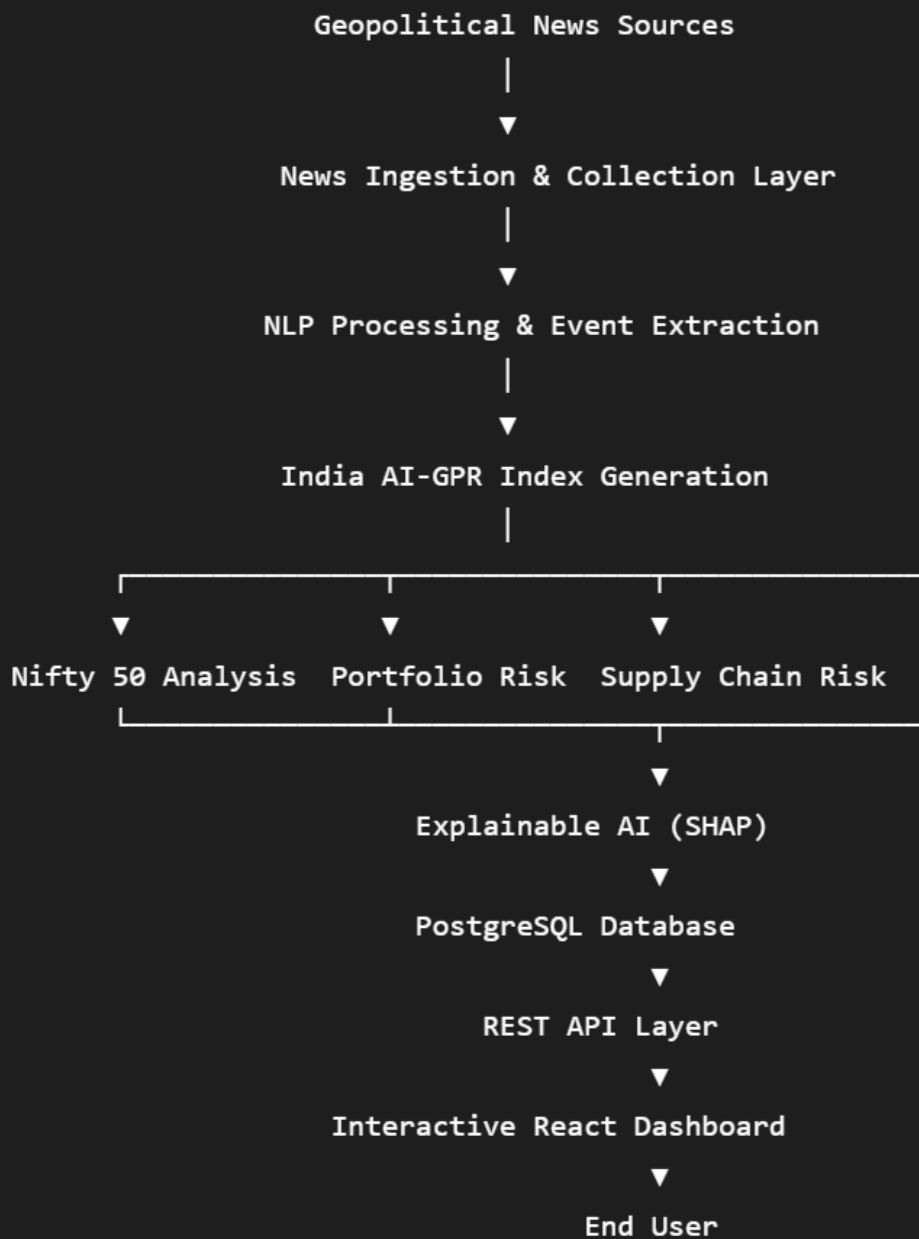


Figure 4.1 Overall System Architecture of Forsyt

Architectural Layers

The architecture of Forsyt is composed of several interconnected layers, each responsible for a specific function within the analytical workflow.

Table 4.1 Architectural Layers of Forsyt

Layer	Primary Responsibility
News Collection Layer	Collects geopolitical news from multiple sources for analysis.
NLP Processing Layer	Performs preprocessing, event extraction, and information structuring using transformer-based NLP models.
India AI-GPR Layer	Generates the India AI-GPR Index from extracted geopolitical events.
Financial Analytics Layer	Performs Nifty 50 analysis, portfolio risk assessment, and supply chain risk evaluation.
Explainable AI Layer	Generates interpretable explanations supporting machine learning outputs.
Data Management Layer	Stores processed events, risk indices, and analytical results within PostgreSQL.
API Layer	Enables communication between backend analytical modules and the frontend interface through REST APIs.
Presentation Layer	Displays analytical insights through an interactive React dashboard.

Architectural Workflow

The execution flow of Forsyt can be summarized as follows:

1. Geopolitical news is collected from configured news sources.
2. News articles undergo preprocessing and NLP-based event extraction.
3. Extracted events are analysed to generate the India AI-GPR Index.
4. The generated geopolitical risk information is supplied to the financial analytics modules.
5. Explainable AI techniques interpret the generated predictions.
6. Processed analytical outputs are stored in the PostgreSQL database.
7. REST APIs retrieve processed information for visualization.
8. The interactive dashboard presents geopolitical intelligence to the end user.

Key Characteristics of the Architecture

The proposed architecture provides several advantages that support the objectives of Forsyt.

- **Modular Design** – Each subsystem performs an independent analytical function, simplifying maintenance and future enhancements.
- **Scalability** – Additional analytical modules and news sources can be incorporated without significant architectural changes.
- **Interoperability** – REST APIs facilitate seamless communication among backend services and the frontend dashboard.
- **Explainability** – SHAP-based explanations improve the transparency and interpretability of predictive analytics.
- **Maintainability** – Separation of concerns allows individual modules to be tested and updated independently.
- **Extensibility** – The architecture supports future integration of additional datasets, predictive models, and visualization modules.

Architecture Summary

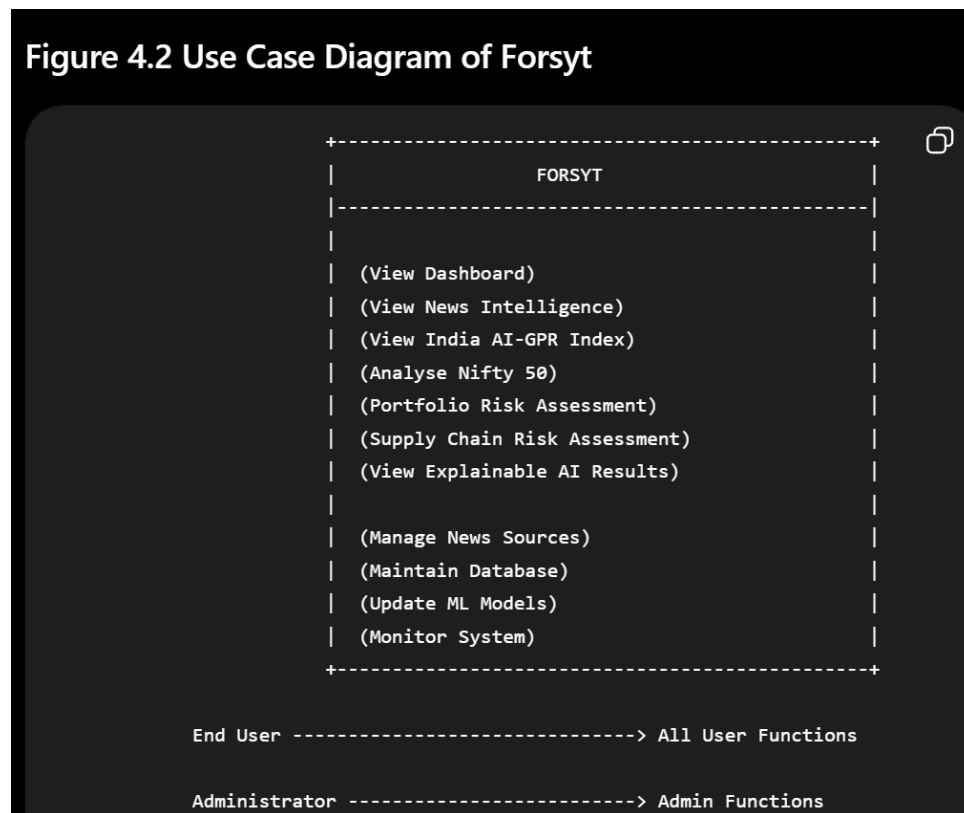
The layered architecture adopted in Forsyt enables the systematic transformation of raw geopolitical news into structured intelligence suitable for financial decision support. By integrating data acquisition, Natural Language Processing, geopolitical risk modelling, machine learning, explainable AI, database management, and interactive visualization within a unified framework, the proposed architecture provides a scalable and maintainable foundation for India-centric geopolitical intelligence. The deployment documentation also reflects this modular design through separate database, pipeline, API, and dashboard services.

4.2 Design Level Diagrams

Design-level diagrams provide a graphical representation of the structure, behaviour, and interaction of the Forsyt platform. These diagrams assist in understanding the overall system design by illustrating how different components communicate, how users interact with the application, and how the software is deployed. The major design diagrams prepared for Forsyt are presented in the following sections.

4.2.1 Use Case Diagram

Figure 4.2 Use Case Diagram of Forsyt



Explanation

The Use Case Diagram provides a high-level functional view of the Forsyt platform by illustrating the interactions between external actors and the services offered by the system. It identifies the major functionalities available to users without describing the internal implementation of the software.

The **End User** interacts with the analytical features of the platform through the interactive dashboard. These functionalities include viewing processed geopolitical news, monitoring the India AI-GPR Index, analysing Nifty 50 trends, assessing portfolio risk, evaluating supply chain risks, and viewing explainable AI results generated by the prediction models.

The **Administrator** is responsible for maintaining the operational aspects of the platform. Administrative activities include managing news sources, maintaining the

application database, updating analytical models, and monitoring the overall health of the system.

Table 4.2 Major Use Cases

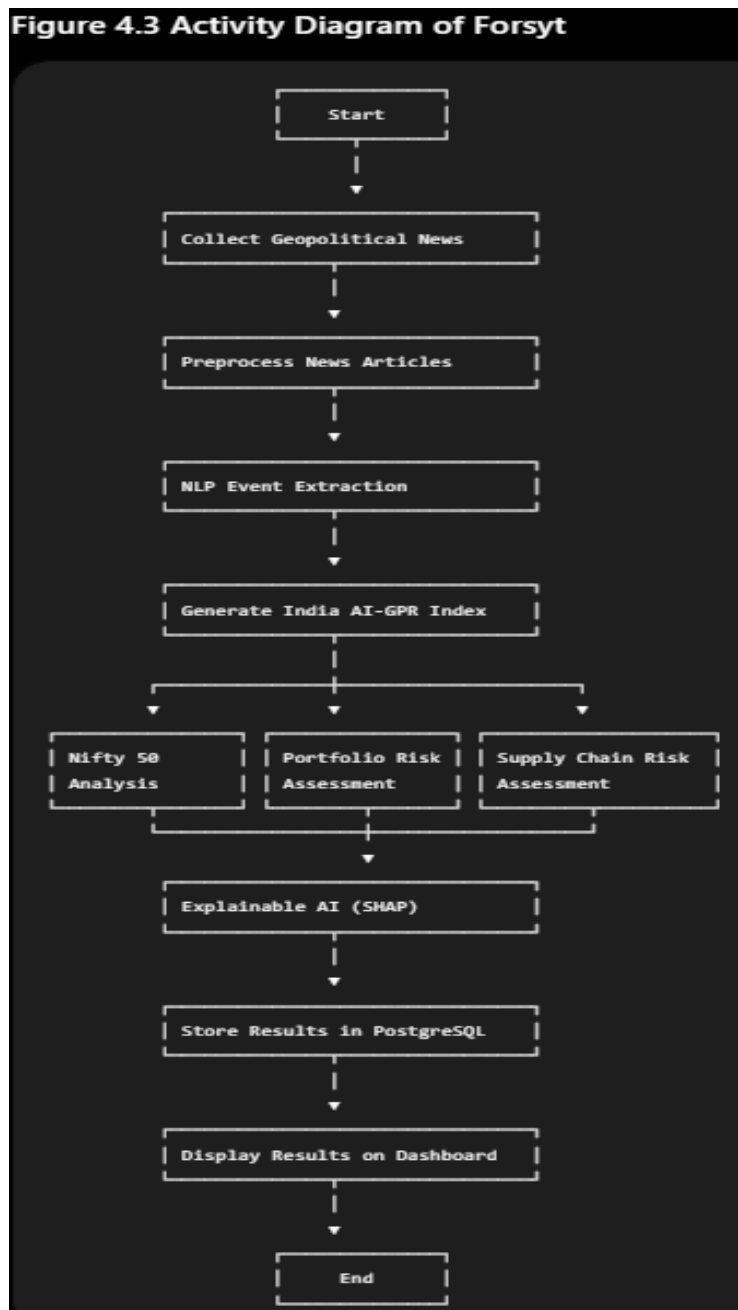
Use Case	Purpose
View Dashboard	Displays consolidated geopolitical intelligence and financial analytics.
View News Intelligence	Displays processed geopolitical news and extracted events.
View India AI-GPR Index	Displays geopolitical risk trends and index values.
Analyse Nifty 50	Evaluates the impact of geopolitical developments on market behaviour.
Portfolio Risk Assessment	Analyses geopolitical exposure of investment portfolios.
Supply Chain Risk Assessment	Evaluates geopolitical risks affecting trade and logistics.
View Explainable AI Results	Displays explanations for model predictions.
Manage News Sources	Maintains configured geopolitical news feeds.
Maintain Database	Manages persistent application data.
Update ML Models	Updates analytical models when required.
Monitor System	Monitors the operational status of the platform.

Significance

The Use Case Diagram defines the functional scope of Forsyt by identifying the primary services provided by the platform and the interactions between users and the system. It serves as the foundation for the behavioural diagrams presented in the following sections.

4.2.2 Activity Diagram

Figure 4.3 Activity Diagram of Forsyt



Explanation

The Activity Diagram illustrates the sequential workflow followed by the Forsyt platform to transform raw geopolitical news into actionable financial intelligence. The process begins with the collection of geopolitical news articles from configured news sources.

The collected articles are then preprocessed to remove unnecessary information and prepare the text for analysis.

The preprocessed news is passed to the Natural Language Processing (NLP) module, where significant geopolitical entities and events are extracted. The structured event data is subsequently used to generate the India AI-GPR Index, which represents the estimated geopolitical risk associated with current developments.

The generated index serves as the input for three analytical modules: **Nifty 50 Analysis**, **Portfolio Risk Assessment**, and **Supply Chain Risk Assessment**. The outputs of these modules are further processed using the Explainable AI component, which provides interpretable explanations for the generated analytical results. Finally, the processed information is stored in the PostgreSQL database and displayed to users through the interactive dashboard.

Table 4.3 Activity Description

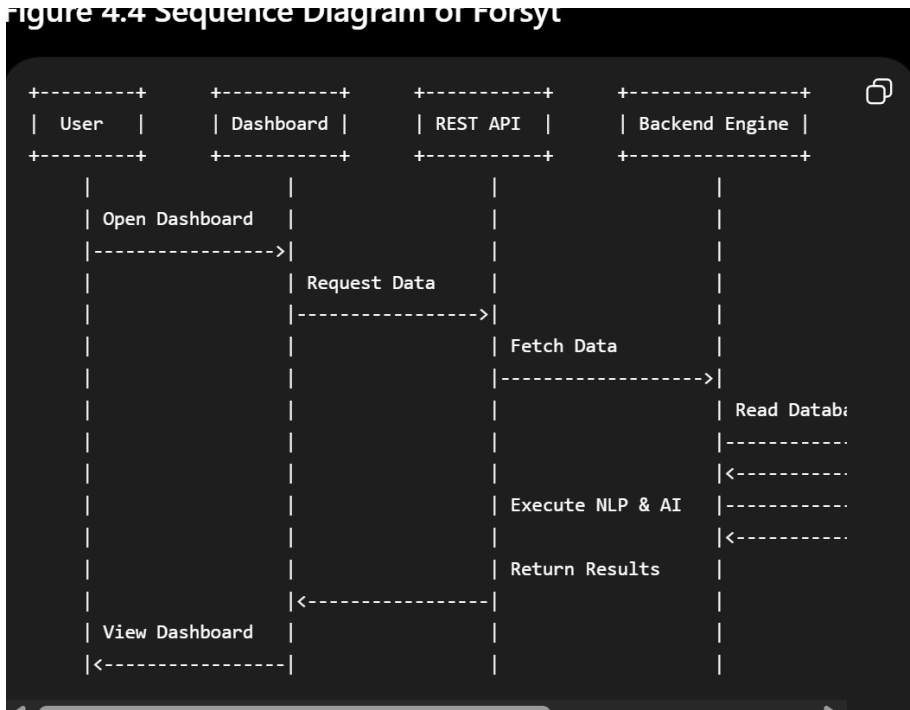
Activity	Purpose
Collect Geopolitical News	Retrieve relevant news articles from configured sources.
Preprocess News	Clean and prepare textual data for analysis.
NLP Event Extraction	Identify important geopolitical entities and events.
Generate India AI-GPR Index	Compute the geopolitical risk indicator from extracted events.
Nifty 50 Analysis	Evaluate the impact of geopolitical events on market behaviour.
Portfolio Risk Assessment	Analyse the geopolitical exposure of investment portfolios.
Supply Chain Risk Assessment	Evaluate risks affecting logistics and trade networks.
Explainable AI	Generate interpretable explanations for analytical outputs.
Store Results	Save processed information in the PostgreSQL database.
Display Dashboard	Present analytical insights to the user through the interface.

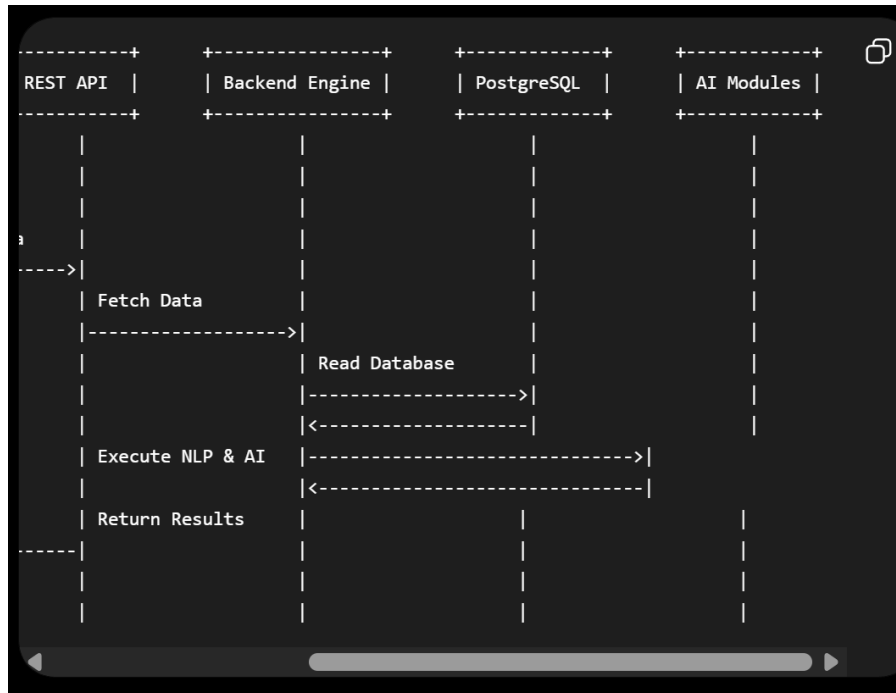
Significance

The Activity Diagram represents the operational workflow of Forsyt by illustrating the sequence of activities performed by the system, from news acquisition to visualization of analytical results. It demonstrates how individual modules are connected to form an integrated analytical pipeline and provides a clear understanding of the overall processing logic adopted by the platform.

4.2.3 Sequence Diagram

Figure 4.4 Sequence Diagram of Forsyt





Explanation

The Sequence Diagram illustrates the chronological interaction among the major software components of Forsyt during the generation and presentation of geopolitical intelligence. The interaction begins when the user accesses the interactive dashboard to view analytical results.

The dashboard sends a request to the REST API, which forwards the request to the backend analytical engine. The backend retrieves previously processed information from the PostgreSQL database and invokes the Artificial Intelligence modules whenever analytical processing is required. These modules perform Natural Language Processing, geopolitical risk estimation, and predictive analysis before returning the processed results to the backend.

The backend consolidates the generated outputs and returns them through the REST API to the dashboard. Finally, the dashboard presents the processed geopolitical intelligence, India AI-GPR Index, financial analysis, and explainable AI results to the user.

Table 4.4 Sequence of Interactions

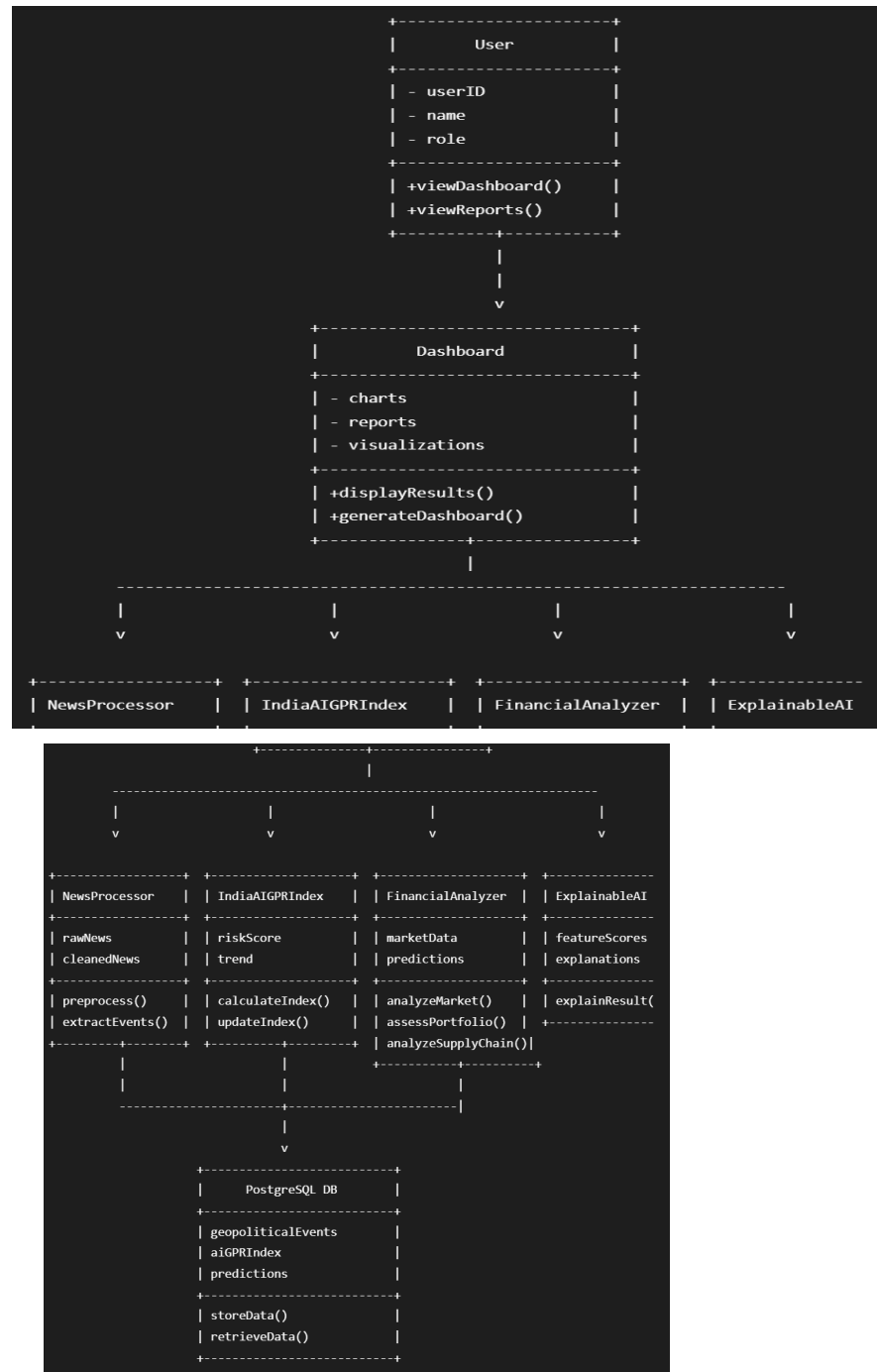
Step	Interaction
1	User opens the Forsyt dashboard.
2	Dashboard sends a request to the REST API.
3	REST API forwards the request to the backend engine.
4	Backend retrieves processed information from PostgreSQL.
5	Backend invokes the AI modules for analytical processing.
6	AI modules return processed analytical outputs.
7	Backend sends the generated results to the REST API.
8	REST API returns the processed data to the dashboard.
9	Dashboard displays the analytical results to the user.

Significance

The Sequence Diagram illustrates the order in which software components communicate during the execution of Forsyt. It highlights the interaction between the user interface, REST API, backend services, database, and AI modules, demonstrating how requests are processed and how analytical results are generated and presented. This diagram provides a clear understanding of the runtime behaviour of the proposed system and complements the structural representation provided by the System Architecture.

4.2.4 Class Diagram

Figure 4.5 Class Diagram of Forsyt



Explanation

The Class Diagram illustrates the static structure of the Forsyt platform by representing the primary software classes, their responsibilities, and their relationships. Each class encapsulates a specific functional responsibility within the application, allowing the system to maintain a modular and maintainable architecture.

The **User** class represents individuals interacting with the application through the dashboard. User requests are handled by the **Dashboard** class, which acts as the presentation layer responsible for displaying analytical outputs.

The **NewsProcessor** class manages geopolitical news collection, preprocessing, and event extraction. The processed information is supplied to the **IndiaAIGPRIndex** class, which computes the geopolitical risk indicator used throughout the platform. The generated index is then utilised by the **FinancialAnalyzer** class to perform Nifty 50 analysis, portfolio risk assessment, and supply chain risk evaluation.

The **ExplainableAI** class generates interpretable explanations for the analytical results, improving transparency and user confidence. Finally, the **PostgreSQL Database** class stores processed events, geopolitical indices, and analytical outputs, ensuring persistent storage and efficient retrieval of information.

Table 4.5 Description of Major Classes

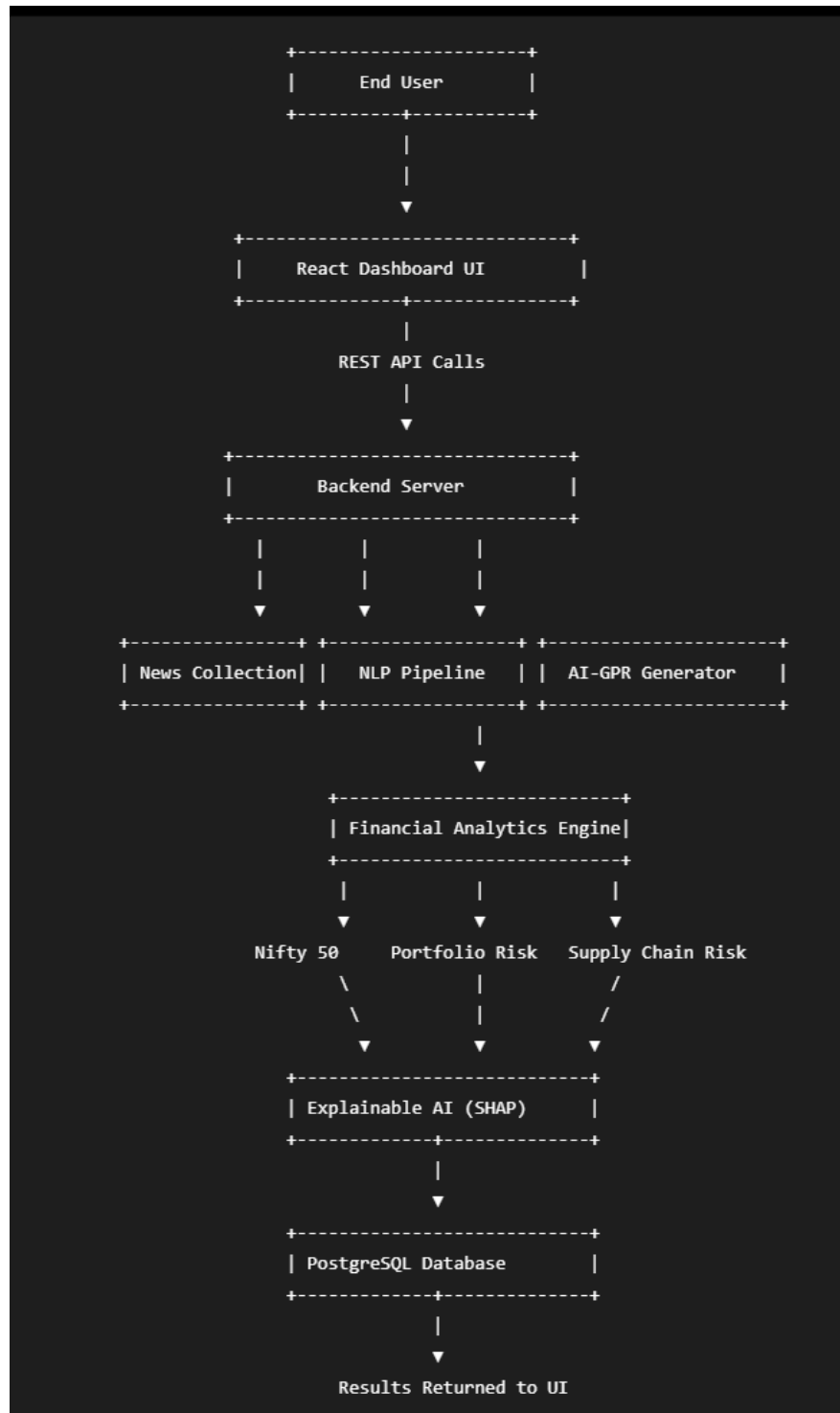
Class	Primary Responsibility
User	Initiates interaction with the platform through the dashboard.
Dashboard	Displays analytical outputs and visualizations.
NewsProcessor	Collects, preprocesses, and extracts geopolitical events from news.
IndiaAIGPRIndex	Generates the India AI-GPR Index from extracted events.
FinancialAnalyzer	Performs market, portfolio, and supply chain analysis.
ExplainableAI	Produces interpretable explanations for model predictions.
PostgreSQL Database	Stores and retrieves processed application data.

Significance

The Class Diagram provides a static representation of the software structure of Forsyt. It demonstrates how the major classes collaborate to process geopolitical news, generate analytical outputs, and present meaningful insights to users. The diagram also highlights the modular organization of the system, which simplifies maintenance, testing, and future enhancements.

4.2.5 Component Diagram

Figure 4.6 Component Diagram of Forsyt



Explanation

The Component Diagram illustrates the high-level software architecture of Forsyt by representing the major software components and the interactions among them. Each component performs a specific responsibility within the analytical workflow while collaborating with other components to generate geopolitical intelligence and financial insights.

The **React Dashboard** serves as the presentation layer through which users interact with the system. User requests are transmitted to the backend through REST APIs. The backend coordinates the execution of the News Collection module, NLP Pipeline, and India AI-GPR Generator to transform raw geopolitical news into structured geopolitical intelligence.

The generated India AI-GPR Index is supplied to the **Financial Analytics Engine**, which performs Nifty 50 analysis, portfolio risk assessment, and supply chain risk assessment. The analytical outputs are further processed by the **Explainable AI (SHAP)** module to generate interpretable explanations before being stored in the PostgreSQL database. The processed results are then retrieved by the backend and presented through the interactive dashboard.

Table 4.6 Description of Software Components

Component	Responsibility
React Dashboard	Provides the graphical user interface for interacting with Forsyt.
Backend Server	Coordinates communication among software modules and processes user requests.
News Collection Module	Retrieves geopolitical news from configured news sources.
NLP Pipeline	Extracts structured geopolitical information from news articles.
India AI-GPR Generator	Computes the India AI-GPR Index from extracted events.
Financial Analytics Engine	Performs market analysis, portfolio risk assessment, and supply chain risk analysis.
Explainable AI Module	Generates explanations supporting predictive analytical results.
PostgreSQL Database	Stores processed events, indices, and analytical outputs.

Significance

The Component Diagram provides a structural representation of the major software modules that constitute the Forsyt platform. It demonstrates how independent components communicate through well-defined interfaces to perform news processing, geopolitical risk analysis, financial analytics, explainable AI, and data management. The modular architecture enhances maintainability, scalability, and future extensibility by allowing individual components to be updated or replaced without affecting the overall system.

4.2.6 Deployment Diagram

Figure 4.7 Deployment Diagram of Forsyt

Figure 4.7 Deployment Diagram of Forsyt



Explanation

The Deployment Diagram illustrates the physical deployment architecture of the Forsyt platform by showing how the software components are distributed across the client and server environments.

The **Client Device** represents the user's system, where the application is accessed through a standard web browser. Users interact with the React-based dashboard to visualize geopolitical intelligence, financial analytics, and AI-generated insights.

The **Application Server** hosts the core functionality of the platform, including the backend services, REST APIs, Natural Language Processing module, India AI-GPR generation module, financial analytics engine, and Explainable AI module. These services process incoming requests, execute analytical workflows, and prepare results for presentation through the dashboard.

The **PostgreSQL Database** serves as the persistent storage layer of the application. It stores processed geopolitical events, India AI-GPR values, financial datasets, prediction results, and other analytical information required by the system. Communication between the application server and the database is performed using SQL queries, while communication between the client and server occurs through HTTP/HTTPS protocols.

Table 4.7 Deployment Components

Deployment Node	Function
Client Device	Provides user access to the Forsyt platform through a web browser.
Web Browser	Displays the interactive dashboard and communicates with the backend.
Application Server	Executes backend logic, NLP processing, AI-GPR generation, and financial analytics.
REST API	Facilitates communication between the frontend and backend services.
PostgreSQL Database	Stores processed data, analytical outputs, and system information.

Significance

The Deployment Diagram provides a physical view of the Forsyt platform by illustrating the hardware and software environments in which the application is executed. It demonstrates the communication between the client interface, application server, and database, highlighting the distributed architecture adopted for implementing the proposed system. This architecture supports scalability, modularity, and efficient management of analytical services.

4.3 User Interface Diagrams

4.3.1 FORSYT Landing Page

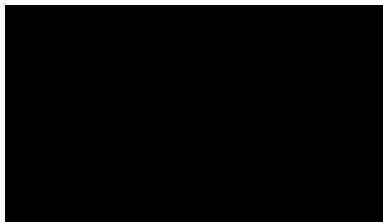


Figure 4.8 FORSYT Landing Page

Description

The FORSYT Landing Page serves as the entry point to the application and introduces users to the platform's geopolitical intelligence capabilities. The interface incorporates the FORSYT branding, an interactive Three.js particle globe, and an overview of the platform's three major intelligence modules. It also presents key India Snapshot KPIs, providing users with an immediate overview of important geopolitical and financial indicators before navigating to the analytical modules.

4.3.2 News Intelligence Interface

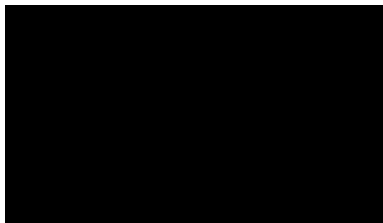


Figure 4.9 News Intelligence Module

Description

The News Intelligence interface provides continuous monitoring of global geopolitical developments relevant to the Indian financial ecosystem. It displays live geopolitical news, risk events, event locations, and intelligence summaries through an interactive interface. Users can apply filters to analyse specific geopolitical events and obtain a structured view of emerging risks.

4.3.3 Indian Macroeconomic Intelligence Interface

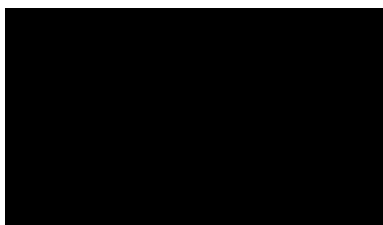


Figure 4.10 Indian Macroeconomic Intelligence Module

Description

This interface presents macroeconomic indicators and financial market information that support geopolitical analysis. The dashboard includes NIFTY 50, Sensex, India VIX, USD/INR exchange rate, market sentiment indicators, FII/DII flows, sector heatmaps, macroeconomic indicators, and an economic calendar. Together, these components provide users with a comprehensive overview of the Indian economic environment.

4.3.4 Trade & Supply Chain Intelligence Interface

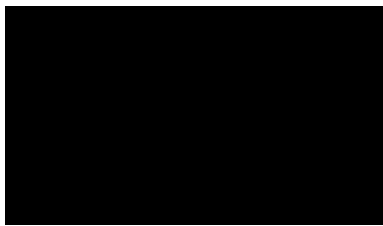


Figure 4.11 Trade & Supply Chain Intelligence Module

Description

The Trade & Supply Chain Intelligence interface analyses the impact of geopolitical developments on global trade corridors and logistics networks. A central interactive corridor map visualizes major strategic trade routes such as IMEC, INSTC, the Red Sea, Suez Canal, Strait of Hormuz, and Strait of Malacca. The interface also presents corridor alerts, trade-route status, supply chain exposure, and operational recommendations for decision-makers.

4.3.5 Portfolio Exposure & GPR Analytics Interface

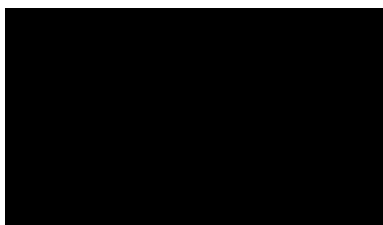


Figure 4.12 Portfolio Exposure & GPR Analytics Module

Description

The Portfolio Exposure & GPR Analytics interface assists users in evaluating the potential financial impact of geopolitical developments on investment portfolios. It presents portfolio allocation, India GPR trends, portfolio risk gauges, sector exposure heatmaps, geopolitical scenario analysis, risk alerts, executive investment briefs, and portfolio recommendations. Interactive visualizations, including a three-dimensional portfolio network, enhance the interpretation of analytical results and support investment decision-making.

Summary

The user interface of FORSYT has been designed to provide an intuitive, interactive, and visually engaging experience for users analysing geopolitical intelligence. Rather than presenting static reports, the platform integrates real-time geopolitical monitoring, macroeconomic indicators, trade and supply chain analysis, and portfolio risk analytics into a unified interface. The use of modern visualization techniques and interactive components improves user engagement while enabling efficient interpretation of complex analytical information.

4.4 Snapshots of Working Prototype

5. CONCLUSION AND FUTURE SCOPE

5.1 Work Accomplished

The development of **Forsyt** successfully achieved the objectives approved during the project's planning and requirement analysis stages. The proposed AI-powered geopolitical intelligence platform was designed, implemented, and validated as an integrated decision-support system for analysing the impact of geopolitical developments on the Indian financial ecosystem. The completed system combines Natural Language Processing (NLP), Artificial Intelligence (AI), financial analytics, explainable AI, and interactive visualization within a unified software platform.

Achievement of Approved Objectives

Table 5.1 Achievement of Project Objectives

Approved Objective	Work Accomplished
Develop an AI-powered geopolitical intelligence platform.	Successfully developed the Forsyt platform integrating AI, NLP, and financial analytics.
Collect and analyse geopolitical news.	Implemented automated news collection and NLP-based event extraction for structured geopolitical intelligence.
Develop an India-centric geopolitical risk indicator.	Designed and implemented the India AI-GPR Index for estimating geopolitical risk.
Analyse the impact of geopolitical events on Indian financial markets.	Developed the Indian Macroeconomic Intelligence module for analysing market indicators and macroeconomic trends.
Assess geopolitical risks affecting trade and supply chains.	Implemented the Trade & Supply Chain Intelligence module for monitoring strategic trade corridors and logistics risks.
Evaluate portfolio exposure to geopolitical developments.	Developed the Portfolio Exposure & GPR Analytics module to support investment risk assessment.
Improve transparency through Explainable AI.	Integrated explainable AI techniques to enhance the interpretability of analytical outputs.
Develop an interactive decision-support interface.	Designed and implemented a modern web-based interface for visualization and analysis.

Major Deliverables

The project successfully delivered the following software components:

- FORSYT Landing Page
- News Intelligence Module
- Indian Macroeconomic Intelligence Module
- Trade & Supply Chain Intelligence Module
- Portfolio Exposure & GPR Analytics Module
- India AI-GPR Index Generation
- NLP-Based Event Extraction Pipeline

- Interactive Dashboard and Visualizations

Overall Outcome

The completed Forsyt platform successfully satisfies the approved project objectives by integrating geopolitical intelligence, macroeconomic analysis, trade and supply chain monitoring, portfolio risk analytics, and explainable AI into a single decision-support platform. The modular architecture, interactive user interface, and AI-driven analytical capabilities provide a scalable foundation for supporting geopolitical risk assessment within the Indian financial ecosystem.

5.2 Conclusions

The successful development of **Forsyt** demonstrates the feasibility of integrating Artificial Intelligence, Natural Language Processing, geopolitical intelligence, and financial analytics into a unified decision-support platform for the Indian financial ecosystem. The project was undertaken to address the limitations of existing geopolitical monitoring systems by providing an India-centric solution capable of analysing geopolitical events and evaluating their potential impact on financial markets, trade corridors, supply chains, and investment portfolios.

Throughout the project, the proposed objectives were successfully achieved by implementing an end-to-end analytical platform that automates the collection of geopolitical news, performs NLP-based event extraction, generates the India AI-GPR Index, and integrates macroeconomic intelligence, trade and supply chain analysis, portfolio exposure assessment, and explainable AI within a single application. The modular architecture adopted during development ensured that individual analytical components could operate efficiently while contributing to an integrated and scalable software solution.

The project also demonstrates the practical application of modern AI technologies in solving real-world analytical problems. By combining automated news intelligence, predictive analytics, and interactive visualization, Forsyt transforms large volumes of unstructured geopolitical information into structured insights that can support informed decision-making. The incorporation of Explainable AI further enhances the transparency and interpretability of analytical outputs, improving user confidence in the generated results.

Overall, Forsyt successfully establishes a comprehensive geopolitical intelligence platform tailored to the Indian financial ecosystem. The completed system provides a strong foundation for future research and development in AI-driven geopolitical risk assessment and demonstrates how emerging technologies can be effectively applied to support strategic financial analysis, investment decision-making, and supply chain risk management in an increasingly interconnected global environment.

5.3 Environmental, Economic, and Social Benefits

The implementation of **Forsyt** extends beyond technological innovation by providing benefits across environmental, economic, and social dimensions. By integrating Artificial Intelligence, Natural Language Processing, and geopolitical intelligence into a unified analytical platform, the system supports informed decision-making while promoting sustainable financial and operational practices.

5.3.1 Environmental Benefits

Forsyt contributes to environmental sustainability by supporting data-driven decision-making that minimizes operational disruptions and resource wastage.

Key environmental benefits include:

- Reduces the need for manual monitoring of multiple geopolitical news sources through automated intelligence gathering.
- Supports efficient supply chain planning by identifying geopolitical risks affecting major trade corridors.
- Enables proactive logistics planning, helping organizations reduce unnecessary transportation disruptions and associated environmental impacts.
- Promotes digital monitoring and reporting, reducing dependence on paper-based documentation and manual reporting processes.
- Encourages efficient utilization of resources through early identification of geopolitical uncertainties.

5.3.2 Economic Benefits

Forsyt provides valuable analytical support for financial institutions, businesses, investors, and policymakers by enabling informed economic decision-making.

Key economic benefits include:

- Supports investment decisions through geopolitical risk assessment.
- Assists in analysing the impact of geopolitical developments on Indian financial markets.
- Improves portfolio risk management by identifying geopolitical exposure.
- Enhances supply chain resilience by monitoring strategic trade routes.
- Provides early geopolitical intelligence that supports proactive business planning.
- Reduces analytical effort by automating data collection, processing, and visualization.

5.3.3 Social Benefits

The platform also contributes to broader societal benefits by improving access to reliable geopolitical intelligence and promoting informed decision-making.

Key social benefits include:

- Improves public understanding of global geopolitical developments through structured analytical insights.
- Supports researchers and students by providing a centralized geopolitical intelligence platform.
- Assists policymakers in analysing the potential impact of geopolitical events on national interests.
- Promotes transparency through Explainable AI, enabling users to understand the reasoning behind analytical outputs.
- Encourages evidence-based decision-making by integrating multiple sources of geopolitical and financial information.
- Enhances awareness of geopolitical risks affecting India's economy and strategic interests.

Summary

Forsyt delivers benefits across environmental, economic, and social dimensions by combining geopolitical intelligence with Artificial Intelligence and financial analytics. The platform enables informed decision-making, improves operational efficiency, strengthens financial risk assessment, and promotes greater awareness of geopolitical developments affecting the Indian financial ecosystem.

5.4 Future Work Plan

Although Forsyt successfully demonstrates the application of Artificial Intelligence for geopolitical intelligence and financial analytics, several enhancements can further improve the platform's capabilities. The following future developments are proposed to enhance scalability, analytical accuracy, and practical usability.

Proposed Future Enhancements

1. Real-Time Data Streaming

Integrate real-time news streaming and financial market data to enable continuous monitoring of geopolitical developments and immediate updates to analytical results.

2. Expanded Geopolitical Coverage

Extend the platform to analyse geopolitical developments across additional countries and regions while maintaining dedicated India-centric analytical capabilities.

3. Enhanced Predictive Analytics

Incorporate advanced forecasting models capable of analysing long-term geopolitical trends and their potential influence on financial markets and economic indicators.

4. Advanced Explainable AI

Enhance the explainability module by providing more detailed visual explanations and interactive analytical reports to improve user understanding of model predictions.

5. Mobile Platform Support

Develop mobile applications to enable users to access geopolitical intelligence, risk assessments, and financial analytics through smartphones and tablets.

6. Additional Financial Indicators

Expand the analytical framework by incorporating additional economic indicators, sector-specific metrics, commodity prices, and other financial datasets to improve decision support.

7. Personalized Decision Support

Implement customizable dashboards and user-specific analytical recommendations based on individual investment preferences and organizational requirements.

8. Continuous Model Improvement

Periodically retrain and validate AI models using newly available geopolitical and financial data to improve analytical accuracy and adaptability to evolving global events.

Summary

The proposed future enhancements aim to improve the scalability, intelligence, and practical applicability of Forsyt. By incorporating real-time analytics, broader geopolitical coverage, enhanced predictive capabilities, mobile accessibility, and continuous model improvement, the platform can evolve into a more comprehensive decision-support system for geopolitical risk analysis and financial intelligence.

REFERENCES

3.1 Investigative Techniques

3. METHODOLOGY ADOPTED

3.1 Investigative Techniques

The development of Forsyt followed a systematic investigative approach to understand how geopolitical events influence the Indian financial ecosystem and how Artificial Intelligence can be used to transform unstructured geopolitical news into actionable financial intelligence. Instead of directly implementing predictive models, the project first investigated existing geopolitical risk measurement techniques, analysed available datasets, evaluated suitable AI models, and validated the proposed methodology against established research.

The investigative process adopted during the development of Forsyt is illustrated in **Figure 3.1**.

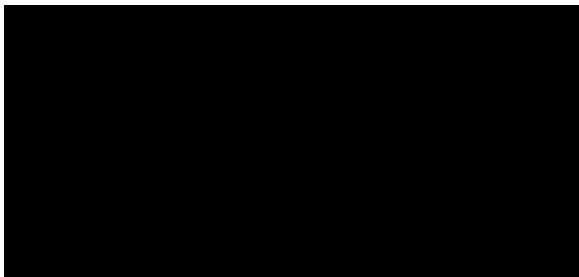


Figure 3.1 Investigative Methodology

(Insert investigative process flowchart here.)

Phases of Investigation

Phase 1: Literature Investigation

The project began with an extensive review of existing research to understand:

- Geopolitical Risk (GPR) Index proposed by Caldara & Iacoviello.
- AI-based Geopolitical Risk (AI-GPR) approaches.
- Economic Policy Uncertainty (EPU).
- Financial market prediction techniques.
- Explainable AI methods.

Outcome: Established the theoretical foundation and identified research gaps.

Phase 2: Problem Investigation

Existing geopolitical monitoring systems and financial prediction models were analysed to identify their limitations.

The investigation revealed that:

- Existing GPR indices are largely global in scope.
- Most systems focus either on news monitoring or financial prediction, but not both.
- Few solutions integrate explainable AI.
- Limited emphasis is placed on the Indian financial ecosystem.

Outcome: Defined the objectives of Forsyt.

Phase 3: Dataset Investigation

Relevant datasets were studied to determine their suitability for geopolitical risk analysis.

The investigation focused on:

- Geopolitical news sources.
- Historical financial market data.
- Data quality.
- Data availability.
- Temporal consistency.

Outcome: Selected datasets for model development and validation.

Phase 4: Model Investigation

Different analytical approaches were evaluated before selecting the final methodology.

The investigation included:

- NLP-based event extraction.
- Transformer-based language models.
- India AI-GPR Index construction.
- Machine learning models for market analysis.
- Explainable AI techniques.

Outcome: Selected the analytical pipeline adopted in Forsyt.

Phase 5: Validation Investigation

The proposed system was evaluated using multiple validation techniques.

These included:

- Comparison with established geopolitical risk methodologies.
- Performance evaluation of NLP models.
- Prediction analysis.
- Explainability assessment.

Outcome: Verified the reliability of the proposed platform.

Summary of Investigative Process

Table 3.1 Investigative Process

Phase	Objective	Outcome
Literature Investigation	Study existing research	Identify theoretical foundation
Problem Investigation	Analyse current systems	Identify research gap
Dataset Investigation	Evaluate available datasets	Select appropriate data
Model Investigation	Compare AI approaches	Design analytical pipeline
Validation Investigation	Evaluate developed system	Verify effectiveness

Justification for the Selected Investigative Technique

The investigative approach adopted for Forsyt combines **research-oriented analysis** with **iterative software development**. This methodology was selected because the project required understanding existing geopolitical risk methodologies before designing an India-centric analytical platform. The phased investigation ensured that every major design decision—from dataset selection to AI model development and validation—was supported by literature, experimental evaluation, and implementation requirements. This

systematic approach improved the reliability, scalability, and interpretability of the final system.

3.2 Proposed Solution

3.2 Proposed Solution

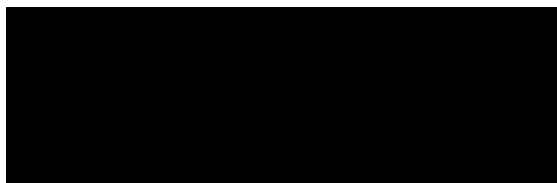
The proposed solution, **Forsyt**, is designed as an AI-powered geopolitical intelligence platform that transforms unstructured geopolitical news into structured analytical insights for the Indian financial ecosystem. The system integrates Natural Language Processing (NLP), Machine Learning (ML), explainable AI, and interactive visualization to analyse geopolitical developments and evaluate their potential impact on financial markets. Unlike conventional geopolitical monitoring platforms that primarily provide descriptive reports, Forsyt combines automated news intelligence, geopolitical risk estimation, financial analytics, and decision-support capabilities within a unified software platform.

The proposed solution follows a modular architecture in which each functional component performs a specialized task while contributing to the overall analytical workflow. Information generated by one module serves as the input for subsequent modules, enabling the system to progressively transform raw geopolitical news into meaningful financial intelligence.

The overall workflow of the proposed solution is illustrated in **Figure 3.2**.

Figure 3.2 Proposed Solution Workflow

(Insert the overall system workflow diagram here.)



Workflow of the Proposed Solution

The proposed solution is implemented through the following sequential stages:

Stage 1: Geopolitical News Acquisition

- Collect geopolitical news from multiple news sources.
- Aggregate articles related to international politics, conflicts, diplomacy, trade, and economic events.
- Store collected articles for further processing.

Output: Raw geopolitical news articles.

Stage 2: News Preprocessing and NLP

- Clean and preprocess collected news articles.
- Remove irrelevant textual information.
- Apply transformer-based Natural Language Processing models.
- Extract meaningful geopolitical entities and events.

Output: Structured geopolitical event data.

Stage 3: India AI-GPR Index Generation

- Analyse extracted geopolitical events.
- Estimate the intensity of geopolitical uncertainty.
- Generate the India AI-GPR Index representing daily geopolitical risk.

Output: India AI-GPR Index.

Stage 4: Financial Analytics

The generated India AI-GPR Index is utilized by multiple analytical modules:

- **Nifty 50 Analysis** – evaluates the relationship between geopolitical developments and market behaviour.
- **Portfolio Risk Advisor** – estimates geopolitical exposure across investment portfolios.
- **Supply Chain Risk Assessment** – analyses geopolitical risks affecting major trade routes and logistics.

Output: Analytical insights for multiple financial applications.

Stage 5: Explainable AI

- Generate explanations supporting machine learning predictions.
- Identify influential features contributing to analytical outputs.

- Improve transparency and interpretability of generated insights.

Output: Explainable analytical results.

Stage 6: Interactive Dashboard

- Present geopolitical events.
- Display India AI-GPR trends.
- Visualize market analysis and risk indicators.
- Provide users with an integrated decision-support interface.

Output: Interactive visualization of analytical results.

Major Functional Modules

Table 3.2 Functional Modules of Forsyt

Module	Primary Function	Output
News Intelligence Module	Collects and preprocesses geopolitical news	Raw news dataset
NLP Event Extraction Module	Extracts structured geopolitical events	Structured event data
India AI-GPR Module	Computes geopolitical risk indicator	India AI-GPR Index
Nifty 50 Analysis Module	Analyses market behaviour	Market insights
Portfolio Risk Advisor	Evaluates geopolitical exposure	Portfolio risk assessment
Supply Chain Risk Module	Assesses logistics and trade risks	Supply chain risk analysis
Explainable AI Module	Explains analytical predictions	Interpretable results
Interactive Dashboard	Presents analytical outputs	User-friendly visualization

Advantages of the Proposed Solution

The proposed solution offers the following advantages over existing approaches:

- Integrates multiple analytical modules within a single platform.
- Combines geopolitical intelligence with financial market analysis.
- Generates an India-centric geopolitical risk indicator.
- Utilizes AI-based Natural Language Processing for event understanding.
- Supports explainable AI to improve transparency and user confidence.
- Provides an interactive dashboard for real-time visualization and analysis.
- Employs a modular architecture that supports future scalability and extension.

Summary

The proposed solution integrates geopolitical news intelligence, Natural Language Processing, India AI-GPR Index generation, financial analytics, explainable AI, and interactive visualization into a unified decision-support platform. The modular design enables efficient interaction among individual components while ensuring scalability, maintainability, and ease of future enhancement. By transforming unstructured geopolitical information into structured analytical intelligence, Forsyt provides users with a comprehensive platform for understanding the potential financial implications of geopolitical developments.

3.3 Work Breakdown Structure

3.3 Work Breakdown Structure

The development of Forsyt was organized into a structured Work Breakdown Structure (WBS) to divide the project into manageable work packages. This approach enabled systematic planning, task allocation, progress monitoring, and integration of individual software components. Each work package represents a major functional area of the project and contributes to the successful implementation of the complete geopolitical intelligence platform.

The overall Work Breakdown Structure of Forsyt is illustrated in **Figure 3.3**.

Figure 3.3 Work Breakdown Structure of Forsyt

(Insert WBS Diagram here.)



Major Work Packages

The project was divided into the following major work packages.

WP1 – Literature Review and Requirement Analysis

Activities

- Study geopolitical risk literature.
- Analyse Caldara's GPR methodology.
- Review AI-GPR approaches.
- Identify system requirements.
- Define project objectives.

Deliverables

- Literature Survey
- Requirement Analysis
- Software Requirement Specification (SRS)

WP2 – Data Collection and Preprocessing

Activities

- Collect geopolitical news.
- Acquire financial market data.
- Clean and preprocess textual data.
- Prepare datasets for analysis.

Deliverables

- Processed news dataset
- Structured financial dataset

WP3 – NLP and Event Extraction Module

Activities

- Perform text preprocessing.
- Extract geopolitical entities.
- Identify significant events.
- Generate structured event information.

Deliverables

- Structured geopolitical events
- NLP processing pipeline

WP4 – India AI-GPR Index Development

Activities

- Analyse extracted geopolitical events.
- Compute geopolitical risk scores.

- Generate India AI-GPR Index.
-

Deliverables

- India AI-GPR Index
- Risk trend analysis

WP5 – Financial Analytics Module

Activities

- Analyse Nifty 50 behaviour.
- Evaluate portfolio exposure.
- Perform supply chain risk analysis.
- Generate analytical insights.

Deliverables

- Market analysis
- Portfolio Risk Advisor
- Supply Chain Risk Assessment

WP6 – Dashboard and System Integration

Activities

- Develop interactive dashboard.
- Integrate backend modules.
- Connect APIs.
- Display analytical outputs.

Deliverables

- Interactive Dashboard
- Integrated software platform

WP7 – Testing and Validation

Activities

- Validate analytical modules.
- Evaluate prediction performance.
- Perform integration testing.
- Debug identified issues.

Deliverables

- Tested software
- Validation results

WP8 – Documentation and Final Deployment

Activities

- Prepare technical documentation.
- Finalize project report.
- Deploy application.
- Demonstrate completed system.

Deliverables

- Technical Report
- Final Software Prototype

Summary of Work Packages

Table 3.3 Work Breakdown Summary

Work Package	Description	Major Deliverable
WP1	Literature Review & Requirement Analysis	SRS and Literature Survey
WP2	Data Collection & Preprocessing	Processed datasets
WP3	NLP Event Extraction	Structured geopolitical events
WP4	India AI-GPR Development	India AI-GPR Index

WP5	Financial Analytics	Risk analysis modules
WP6	Dashboard & Integration	Interactive Dashboard
WP7	Testing & Validation	Verified system
WP8	Documentation & Deployment	Final software and report

Discussion

The Work Breakdown Structure ensured that the project was developed incrementally, with each work package producing clearly defined deliverables that supported the subsequent phase of development. The modular organization also enabled parallel development of analytical components, simplified testing and integration, and improved overall project management. By dividing the project into independent yet interconnected work packages, the development team was able to systematically progress from literature review and data preparation to implementation, validation, and deployment of the complete Forsyt platform.

3.4 Tools and Technology

3.4 Tools and Technology

The successful implementation of Forsyt required the integration of multiple software technologies for application development, data processing, Natural Language Processing (NLP), machine learning, database management, and visualization. The selected technologies were chosen based on their compatibility, scalability, and ability to support the analytical requirements of an AI-powered geopolitical intelligence platform. Each technology contributed to a specific stage of the system's implementation, from data acquisition to interactive visualization.

Technology Stack Used in Forsyt

Table 3.4 Technology Stack

Technology	Role in Implementation
Python	Core programming language used for implementing backend services, data processing, machine learning, and NLP workflows.
React.js	Developed the interactive web interface used for displaying geopolitical insights and analytical outputs.
PostgreSQL	Stored processed geopolitical events, analytical results, and application data.
REST APIs	Enabled communication between frontend and backend components for data exchange.
Hugging Face Transformers	Implemented transformer-based Natural Language Processing for geopolitical news analysis and event extraction.
XGBoost	Developed predictive models for analysing the relationship between geopolitical developments and financial markets.
Scikit-learn	Supported preprocessing, feature engineering, model evaluation, and auxiliary machine learning tasks.
PyTorch	Provided the deep learning framework required for transformer-based NLP models.
SHAP	Generated explainable AI outputs to improve transparency and interpretability of predictive models.

Recharts / D3.js	Created interactive visualizations for geopolitical trends, the India AI-GPR Index, portfolio analytics, and financial market indicators.
Tailwind CSS	Developed a responsive and visually consistent user interface for the dashboard.
Git & GitHub	Managed source code, version control, and collaborative software development.

Technology Usage During Development

The implementation of Forsyt was divided into multiple functional layers, with each technology supporting a specific aspect of the software architecture.

Backend Development

- Python served as the primary backend programming language.
- REST APIs enabled communication between software modules.
- PostgreSQL managed persistent storage of processed data.

Artificial Intelligence and NLP

- Hugging Face Transformers processed geopolitical news articles.
- PyTorch supported transformer-based model execution.
- Scikit-learn handled preprocessing and model evaluation.
- XGBoost implemented predictive analytical models.
- SHAP generated explanations for machine learning predictions.

Frontend and Visualization

- React.js implemented the web-based user interface.
- Recharts / D3.js provided interactive charts and visualizations for geopolitical trends, the India AI-GPR Index, and financial market analytics.
- Tailwind CSS enabled responsive styling and a consistent user interface across dashboard components.

Software Development and Collaboration

- Git tracked software revisions throughout development.
- GitHub facilitated collaborative coding, repository management, and documentation.

Technology Integration

The technologies were integrated to create a modular software architecture. News data collected by the backend is processed using Natural Language Processing models before being analysed by machine learning components. The processed results are stored in PostgreSQL and exposed through REST APIs. The frontend dashboard retrieves these results and presents them using interactive visualizations, enabling users to analyse geopolitical developments and their potential financial implications through a unified interface.

Advantages of the Selected Technology Stack

The adopted technology stack provides several advantages:

- Modular and scalable software architecture.
- Efficient processing of large volumes of geopolitical news.
- Reliable storage and retrieval of analytical data.
- Interactive visualization of geopolitical and financial insights.
- Explainable AI for improved transparency.
- Ease of maintenance through version control and collaborative development.

4.1 System Architecture

4. DESIGN SPECIFICATIONS

4.1 System Architecture

The architecture of Forsyt is designed as a modular, scalable, and AI-driven analytical framework that converts unstructured geopolitical news into structured financial intelligence for the Indian market. Rather than functioning as a conventional news aggregation platform, the system integrates automated data acquisition, Natural Language Processing (NLP), geopolitical risk modelling, machine learning, explainable AI, and interactive visualization within a unified software ecosystem. This modular architecture allows individual components to perform specialized analytical tasks while collectively supporting the end-to-end generation of geopolitical intelligence. The repository also documents this layered implementation approach through separate ingestion, NLP, machine learning, API, dashboard, and deployment components.

The architecture follows a layered processing pipeline in which raw geopolitical news is progressively transformed into structured, interpretable, and actionable information. Information collected from news sources undergoes preprocessing and NLP-based event extraction before being used for India AI-GPR Index generation. The generated geopolitical risk information subsequently supports multiple downstream analytical modules, including Nifty 50 volatility analysis, Portfolio Risk Advisor, and Supply Chain Risk Assessment. The processed analytical outputs are finally delivered to users through an interactive dashboard supported by REST APIs and a PostgreSQL database.

Overall System Architecture

The overall architecture of Forsyt is illustrated in **Figure 4.1**.

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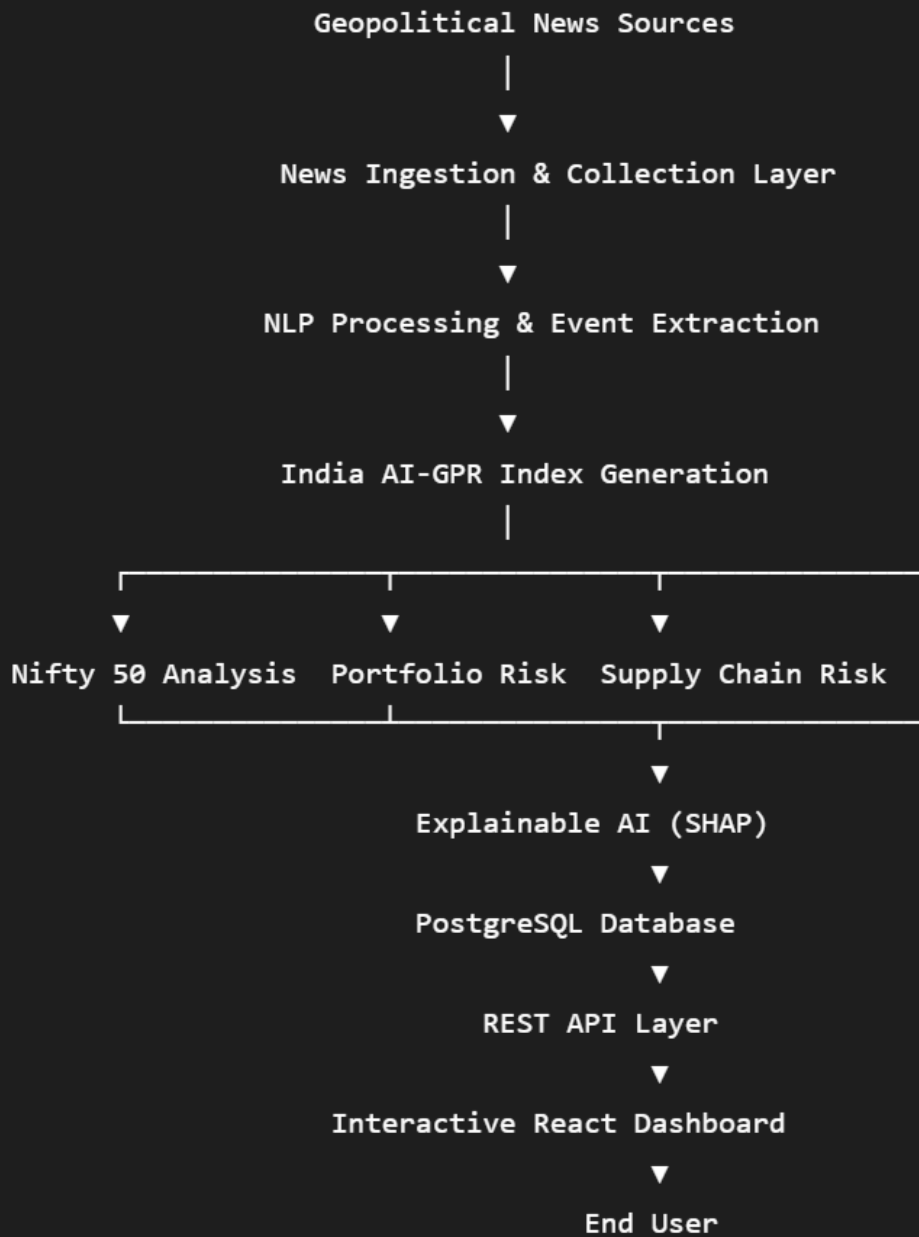


Figure 4.1 Overall System Architecture of Forsyt

(Insert Architecture Diagram here)

Architectural Layers

The architecture of Forsyt is composed of several interconnected layers, each responsible for a specific function within the analytical workflow.

Table 4.1 Architectural Layers of Forsyt

Layer	Primary Responsibility
News Collection Layer	Collects geopolitical news from multiple sources for analysis.
NLP Processing Layer	Performs preprocessing, event extraction, and information structuring using transformer-based NLP models.
India AI-GPR Layer	Generates the India AI-GPR Index from extracted geopolitical events.
Financial Analytics Layer	Performs Nifty 50 analysis, portfolio risk assessment, and supply chain risk evaluation.
Explainable AI Layer	Generates interpretable explanations supporting machine learning outputs.
Data Management Layer	Stores processed events, risk indices, and analytical results within PostgreSQL.
API Layer	Enables communication between backend analytical modules and the frontend interface through REST APIs.
Presentation Layer	Displays analytical insights through an interactive React dashboard.

Architectural Workflow

The execution flow of Forsyt can be summarized as follows:

1. Geopolitical news is collected from configured news sources.
2. News articles undergo preprocessing and NLP-based event extraction.
3. Extracted events are analysed to generate the India AI-GPR Index.
4. The generated geopolitical risk information is supplied to the financial analytics modules.
5. Explainable AI techniques interpret the generated predictions.
6. Processed analytical outputs are stored in the PostgreSQL database.

7. REST APIs retrieve processed information for visualization.
8. The interactive dashboard presents geopolitical intelligence to the end user.

Key Characteristics of the Architecture

The proposed architecture provides several advantages that support the objectives of Forsyt.

- **Modular Design** – Each subsystem performs an independent analytical function, simplifying maintenance and future enhancements.
- **Scalability** – Additional analytical modules and news sources can be incorporated without significant architectural changes.
- **Interoperability** – REST APIs facilitate seamless communication among backend services and the frontend dashboard.
- **Explainability** – SHAP-based explanations improve the transparency and interpretability of predictive analytics.
- **Maintainability** – Separation of concerns allows individual modules to be tested and updated independently.
- **Extensibility** – The architecture supports future integration of additional datasets, predictive models, and visualization modules.

Architecture Summary

The layered architecture adopted in Forsyt enables the systematic transformation of raw geopolitical news into structured intelligence suitable for financial decision support. By integrating data acquisition, Natural Language Processing, geopolitical risk modelling, machine learning, explainable AI, database management, and interactive visualization within a unified framework, the proposed architecture provides a scalable and maintainable foundation for India-centric geopolitical intelligence. The deployment documentation also reflects this modular design through separate database, pipeline, API, and dashboard services.

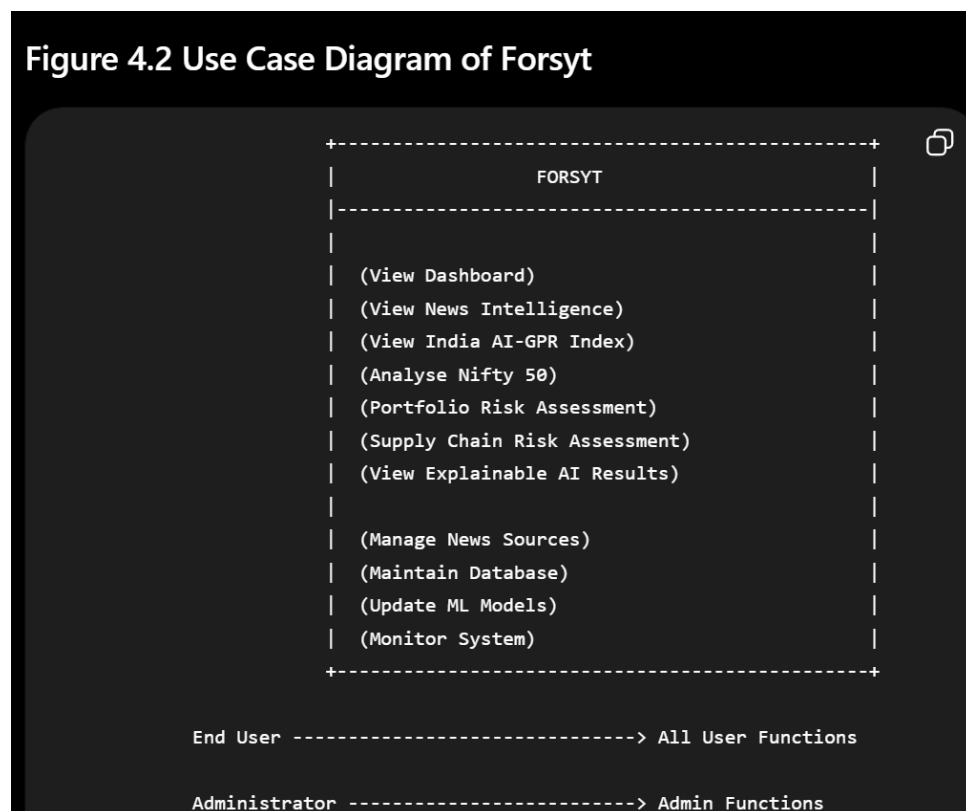
4.2 Design Level Diagrams

4.2 Design Level Diagrams

Design-level diagrams provide a graphical representation of the structure, behaviour, and interaction of the Forsyt platform. These diagrams assist in understanding the overall system design by illustrating how different components communicate, how users interact with the application, and how the software is deployed. The major design diagrams prepared for Forsyt are presented in the following sections.

4.2.1 Use Case Diagram

Figure 4.2 Use Case Diagram of Forsyt



Explanation

The Use Case Diagram provides a high-level functional view of the Forsyt platform by illustrating the interactions between external actors and the services offered by the system. It identifies the major functionalities available to users without describing the internal implementation of the software.

The **End User** interacts with the analytical features of the platform through the interactive dashboard. These functionalities include viewing processed geopolitical news, monitoring the India AI-GPR Index, analysing Nifty 50 trends, assessing portfolio risk, evaluating supply chain risks, and viewing explainable AI results generated by the prediction models.

The **Administrator** is responsible for maintaining the operational aspects of the platform. Administrative activities include managing news sources, maintaining the application database, updating analytical models, and monitoring the overall health of the system.

Table 4.2 Major Use Cases

Use Case	Purpose
View Dashboard	Displays consolidated geopolitical intelligence and financial analytics.
View News Intelligence	Displays processed geopolitical news and extracted events.
View India AI-GPR Index	Displays geopolitical risk trends and index values.
Analyse Nifty 50	Evaluates the impact of geopolitical developments on market behaviour.
Portfolio Risk Assessment	Analyses geopolitical exposure of investment portfolios.
Supply Chain Risk Assessment	Evaluates geopolitical risks affecting trade and logistics.
View Explainable AI Results	Displays explanations for model predictions.
Manage News Sources	Maintains configured geopolitical news feeds.
Maintain Database	Manages persistent application data.

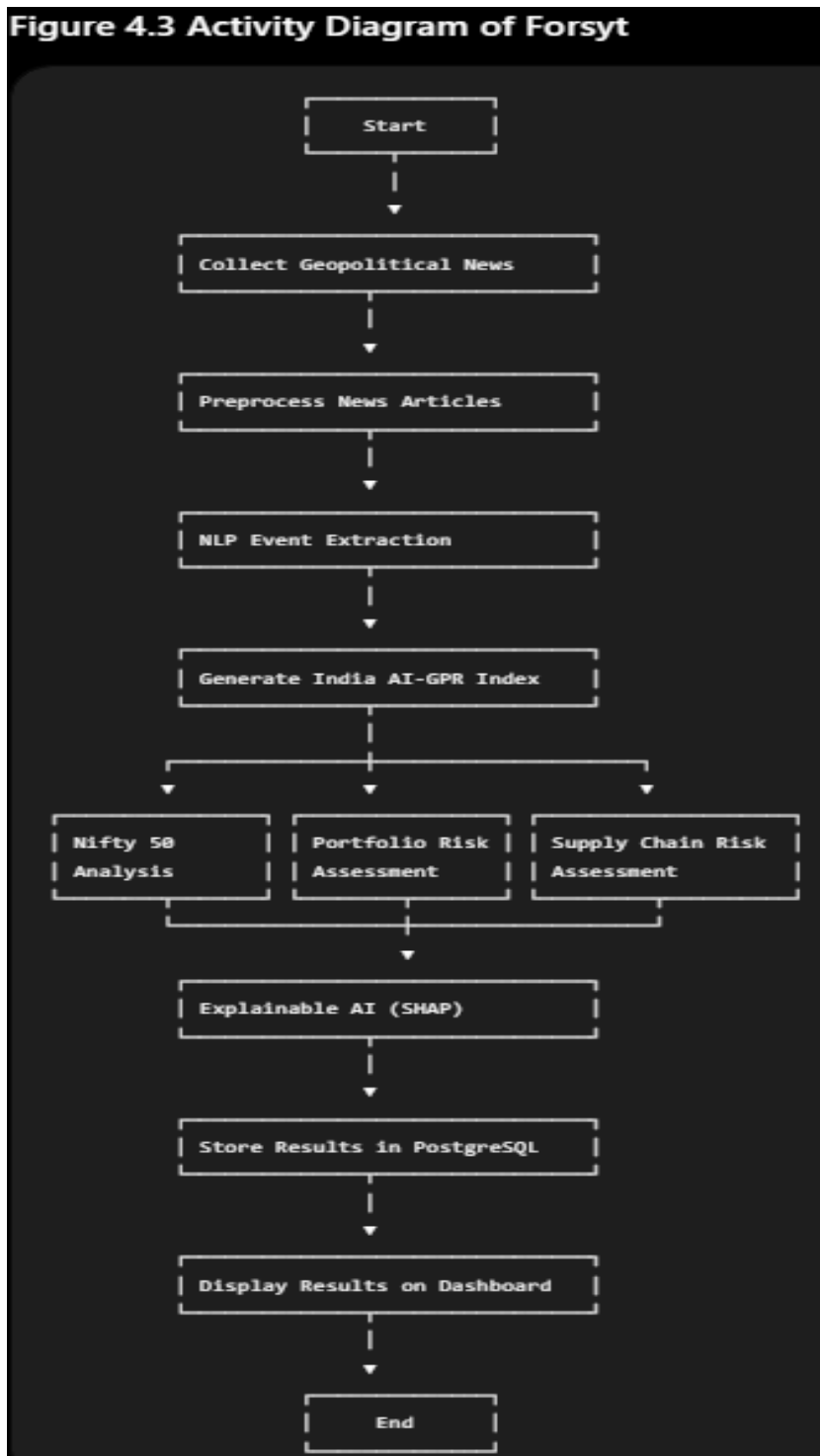
Update ML Models	Updates analytical models when required.
Monitor System	Monitors the operational status of the platform.

Significance

The Use Case Diagram defines the functional scope of Forsyt by identifying the primary services provided by the platform and the interactions between users and the system. It serves as the foundation for the behavioural diagrams presented in the following sections.

4.2.2 Activity Diagram

Figure 4.3 Activity Diagram of Forsyt



Explanation

The Activity Diagram illustrates the sequential workflow followed by the Forsyt platform to transform raw geopolitical news into actionable financial intelligence. The process begins with the collection of geopolitical news articles from configured news sources. The collected articles are then preprocessed to remove unnecessary information and prepare the text for analysis.

The preprocessed news is passed to the Natural Language Processing (NLP) module, where significant geopolitical entities and events are extracted. The structured event data is subsequently used to generate the India AI-GPR Index, which represents the estimated geopolitical risk associated with current developments.

The generated index serves as the input for three analytical modules: **Nifty 50 Analysis**, **Portfolio Risk Assessment**, and **Supply Chain Risk Assessment**. The outputs of these modules are further processed using the Explainable AI component, which provides interpretable explanations for the generated analytical results. Finally, the processed information is stored in the PostgreSQL database and displayed to users through the interactive dashboard.

Table 4.3 Activity Description

Activity	Purpose
Collect Geopolitical News	Retrieve relevant news articles from configured sources.
Preprocess News	Clean and prepare textual data for analysis.
NLP Event Extraction	Identify important geopolitical entities and events.
Generate India AI-GPR Index	Compute the geopolitical risk indicator from extracted events.
Nifty 50 Analysis	Evaluate the impact of geopolitical events on market behaviour.
Portfolio Risk Assessment	Analyse the geopolitical exposure of investment portfolios.
Supply Chain Risk Assessment	Evaluate risks affecting logistics and trade networks.
Explainable AI	Generate interpretable explanations for analytical outputs.
Store Results	Save processed information in the PostgreSQL database.

Display Dashboard	Present analytical insights to the user through the interface.
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Significance

The Activity Diagram represents the operational workflow of Forsyt by illustrating the sequence of activities performed by the system, from news acquisition to visualization of analytical results. It demonstrates how individual modules are connected to form an integrated analytical pipeline and provides a clear understanding of the overall processing logic adopted by the platform.

4.2.3 Sequence Diagram

Figure 4.4 Sequence Diagram of Forsyt



Explanation

The Sequence Diagram illustrates the chronological interaction among the major software components of Forsyt during the generation and presentation of geopolitical intelligence. The interaction begins when the user accesses the interactive dashboard to view analytical results.

The dashboard sends a request to the REST API, which forwards the request to the backend analytical engine. The backend retrieves previously processed information from the PostgreSQL database and invokes the Artificial Intelligence modules whenever analytical processing is required. These modules perform Natural Language Processing, geopolitical risk estimation, and predictive analysis before returning the processed results to the backend.

The backend consolidates the generated outputs and returns them through the REST API to the dashboard. Finally, the dashboard presents the processed geopolitical intelligence, India AI-GPR Index, financial analysis, and explainable AI results to the user.

Table 4.4 Sequence of Interactions

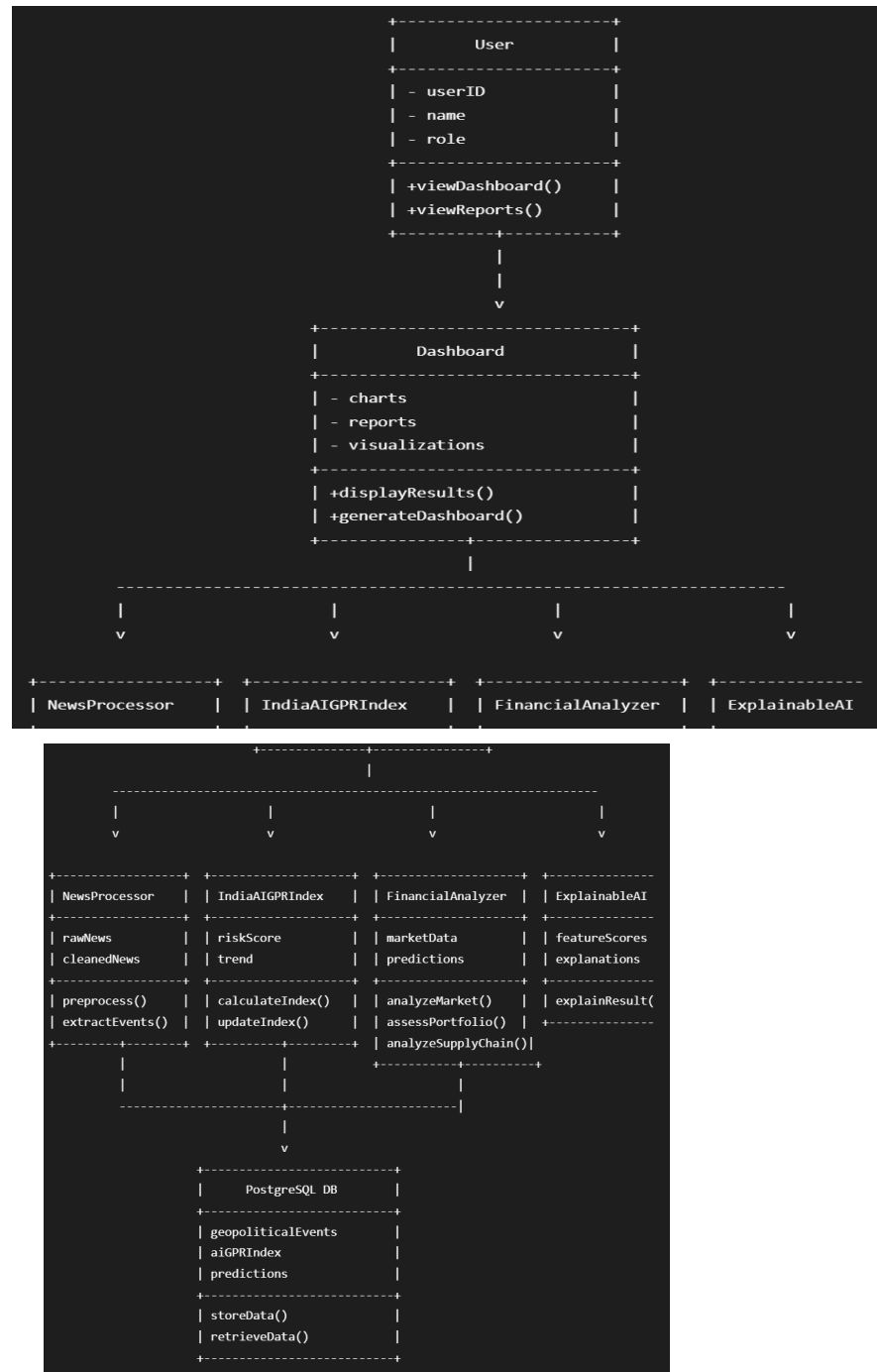
Step	Interaction
1	User opens the Forsyt dashboard.
2	Dashboard sends a request to the REST API.
3	REST API forwards the request to the backend engine.
4	Backend retrieves processed information from PostgreSQL.
5	Backend invokes the AI modules for analytical processing.
6	AI modules return processed analytical outputs.
7	Backend sends the generated results to the REST API.
8	REST API returns the processed data to the dashboard.
9	Dashboard displays the analytical results to the user.

Significance

The Sequence Diagram illustrates the order in which software components communicate during the execution of Forsyt. It highlights the interaction between the user interface, REST API, backend services, database, and AI modules, demonstrating how requests are processed and how analytical results are generated and presented. This diagram provides a clear understanding of the runtime behaviour of the proposed system and complements the structural representation provided by the System Architecture.

4.2.4 Class Diagram

Figure 4.5 Class Diagram of Forsyt



Explanation

The Class Diagram illustrates the static structure of the Forsyt platform by representing the primary software classes, their responsibilities, and their relationships. Each class encapsulates a specific functional responsibility within the application, allowing the system to maintain a modular and maintainable architecture.

The **User** class represents individuals interacting with the application through the dashboard. User requests are handled by the **Dashboard** class, which acts as the presentation layer responsible for displaying analytical outputs.

The **NewsProcessor** class manages geopolitical news collection, preprocessing, and event extraction. The processed information is supplied to the **IndiaAIGPRIndex** class, which computes the geopolitical risk indicator used throughout the platform. The generated index is then utilised by the **FinancialAnalyzer** class to perform Nifty 50 analysis, portfolio risk assessment, and supply chain risk evaluation.

The **ExplainableAI** class generates interpretable explanations for the analytical results, improving transparency and user confidence. Finally, the **PostgreSQL Database** class stores processed events, geopolitical indices, and analytical outputs, ensuring persistent storage and efficient retrieval of information.

Table 4.5 Description of Major Classes

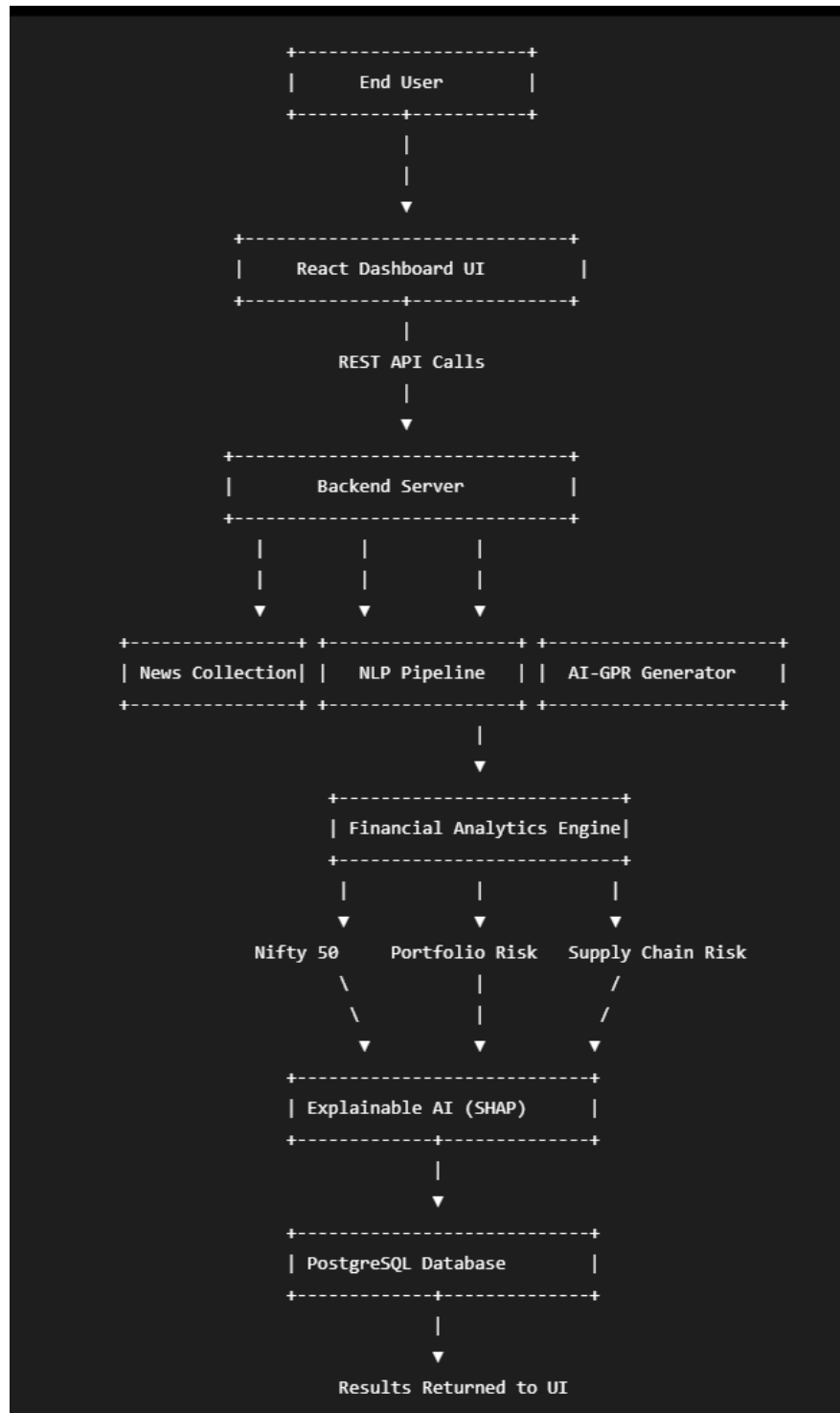
Class	Primary Responsibility
User	Initiates interaction with the platform through the dashboard.
Dashboard	Displays analytical outputs and visualizations.
NewsProcessor	Collects, preprocesses, and extracts geopolitical events from news.
IndiaAIGPRIndex	Generates the India AI-GPR Index from extracted events.
FinancialAnalyzer	Performs market, portfolio, and supply chain analysis.
ExplainableAI	Produces interpretable explanations for model predictions.
PostgreSQL Database	Stores and retrieves processed application data.

Significance

The Class Diagram provides a static representation of the software structure of Forsyt. It demonstrates how the major classes collaborate to process geopolitical news, generate analytical outputs, and present meaningful insights to users. The diagram also highlights the modular organization of the system, which simplifies maintenance, testing, and future enhancements.

4.2.5 Component Diagram

Figure 4.6 Component Diagram of Forsyt



Explanation

The Component Diagram illustrates the high-level software architecture of Forsyt by representing the major software components and the interactions among them. Each component performs a specific responsibility within the analytical workflow while collaborating with other components to generate geopolitical intelligence and financial insights.

The **React Dashboard** serves as the presentation layer through which users interact with the system. User requests are transmitted to the backend through REST APIs. The backend coordinates the execution of the News Collection module, NLP Pipeline, and India AI-GPR Generator to transform raw geopolitical news into structured geopolitical intelligence.

The generated India AI-GPR Index is supplied to the **Financial Analytics Engine**, which performs Nifty 50 analysis, portfolio risk assessment, and supply chain risk assessment. The analytical outputs are further processed by the **Explainable AI (SHAP)** module to generate interpretable explanations before being stored in the PostgreSQL database. The processed results are then retrieved by the backend and presented through the interactive dashboard.

Table 4.6 Description of Software Components

Component	Responsibility
React Dashboard	Provides the graphical user interface for interacting with Forsyt.
Backend Server	Coordinates communication among software modules and processes user requests.
News Collection Module	Retrieves geopolitical news from configured news sources.
NLP Pipeline	Extracts structured geopolitical information from news articles.
India AI-GPR Generator	Computes the India AI-GPR Index from extracted events.
Financial Analytics Engine	Performs market analysis, portfolio risk assessment, and supply chain risk analysis.
Explainable AI Module	Generates explanations supporting predictive analytical results.
PostgreSQL Database	Stores processed events, indices, and analytical outputs.

Significance

The Component Diagram provides a structural representation of the major software modules that constitute the Forsyt platform. It demonstrates how independent components communicate through well-defined interfaces to perform news processing, geopolitical risk analysis, financial analytics, explainable AI, and data management. The modular architecture enhances maintainability, scalability, and future extensibility by allowing individual components to be updated or replaced without affecting the overall system.

4.2.6 Deployment Diagram

Figure 4.7 Deployment Diagram of Forsyt

Figure 4.7 Deployment Diagram of Forsyt



Explanation

The Deployment Diagram illustrates the physical deployment architecture of the Forsyt platform by showing how the software components are distributed across the client and server environments.

The **Client Device** represents the user's system, where the application is accessed through a standard web browser. Users interact with the React-based dashboard to visualize geopolitical intelligence, financial analytics, and AI-generated insights.

The **Application Server** hosts the core functionality of the platform, including the backend services, REST APIs, Natural Language Processing module, India AI-GPR generation module, financial analytics engine, and Explainable AI module. These services process incoming requests, execute analytical workflows, and prepare results for presentation through the dashboard.

The **PostgreSQL Database** serves as the persistent storage layer of the application. It stores processed geopolitical events, India AI-GPR values, financial datasets, prediction results, and other analytical information required by the system. Communication between the application server and the database is performed using SQL queries, while communication between the client and server occurs through HTTP/HTTPS protocols.

Table 4.7 Deployment Components

Deployment Node	Function
Client Device	Provides user access to the Forsyt platform through a web browser.
Web Browser	Displays the interactive dashboard and communicates with the backend.
Application Server	Executes backend logic, NLP processing, AI-GPR generation, and financial analytics.
REST API	Facilitates communication between the frontend and backend services.
PostgreSQL Database	Stores processed data, analytical outputs, and system information.

Significance

The Deployment Diagram provides a physical view of the Forsyt platform by illustrating the hardware and software environments in which the application is executed. It demonstrates the communication between the client interface, application server, and database, highlighting the distributed architecture adopted for implementing the proposed system. This architecture supports scalability, modularity, and efficient management of analytical services.

4.3 User Interface Diagrams

4.3 User Interface Diagrams

4.3.1 FORSYT Landing Page

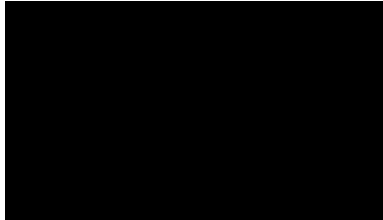


Figure 4.8 FORSYT Landing Page

Description

The FORSYT Landing Page serves as the entry point to the application and introduces users to the platform's geopolitical intelligence capabilities. The interface incorporates the FORSYT branding, an interactive Three.js particle globe, and an overview of the platform's three major intelligence modules. It also presents key India Snapshot KPIs, providing users with an immediate overview of important geopolitical and financial indicators before navigating to the analytical modules.

4.3.2 News Intelligence Interface

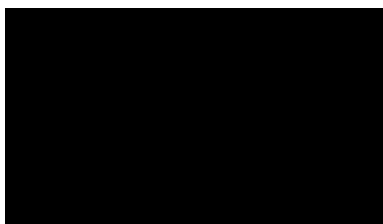


Figure 4.9 News Intelligence Module

Description

The News Intelligence interface provides continuous monitoring of global geopolitical developments relevant to the Indian financial ecosystem. It displays live geopolitical news, risk events, event locations, and intelligence summaries through an interactive

interface. Users can apply filters to analyse specific geopolitical events and obtain a structured view of emerging risks.

4.3.3 Indian Macroeconomic Intelligence Interface

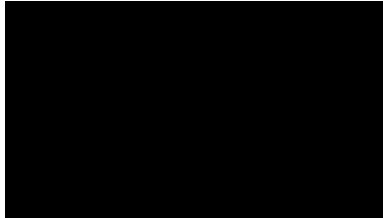


Figure 4.10 Indian Macroeconomic Intelligence Module

Description

This interface presents macroeconomic indicators and financial market information that support geopolitical analysis. The dashboard includes NIFTY 50, Sensex, India VIX, USD/INR exchange rate, market sentiment indicators, FII/DII flows, sector heatmaps, macroeconomic indicators, and an economic calendar. Together, these components provide users with a comprehensive overview of the Indian economic environment.

4.3.4 Trade & Supply Chain Intelligence Interface

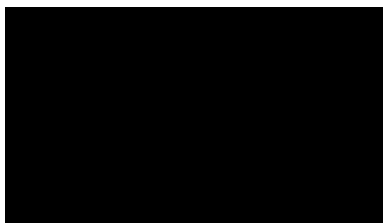


Figure 4.11 Trade & Supply Chain Intelligence Module

Description

The Trade & Supply Chain Intelligence interface analyses the impact of geopolitical developments on global trade corridors and logistics networks. A central interactive corridor map visualizes major strategic trade routes such as IMEC, INSTC, the Red Sea, Suez Canal, Strait of Hormuz, and Strait of Malacca. The interface also presents

corridor alerts, trade-route status, supply chain exposure, and operational recommendations for decision-makers.

4.3.5 Portfolio Exposure & GPR Analytics Interface

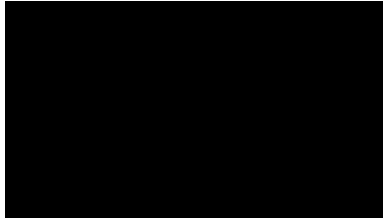


Figure 4.12 Portfolio Exposure & GPR Analytics Module

Description

The Portfolio Exposure & GPR Analytics interface assists users in evaluating the potential financial impact of geopolitical developments on investment portfolios. It presents portfolio allocation, India GPR trends, portfolio risk gauges, sector exposure heatmaps, geopolitical scenario analysis, risk alerts, executive investment briefs, and portfolio recommendations. Interactive visualizations, including a three-dimensional portfolio network, enhance the interpretation of analytical results and support investment decision-making.

Summary

The user interface of FORSYT has been designed to provide an intuitive, interactive, and visually engaging experience for users analysing geopolitical intelligence. Rather than presenting static reports, the platform integrates real-time geopolitical monitoring, macroeconomic indicators, trade and supply chain analysis, and portfolio risk analytics into a unified interface. The use of modern visualization techniques and interactive components improves user engagement while enabling efficient interpretation of complex analytical information.

4.4 Snapshots of Working Prototype

4.4 Snapshots of Working Prototype

5.1 Work Accomplished

5. CONCLUSION AND FUTURE SCOPE

5.1 Work Accomplished

The development of **Forsyt** successfully achieved the objectives approved during the project's planning and requirement analysis stages. The proposed AI-powered geopolitical intelligence platform was designed, implemented, and validated as an integrated decision-support system for analysing the impact of geopolitical developments on the Indian financial ecosystem. The completed system combines Natural Language Processing (NLP), Artificial Intelligence (AI), financial analytics, explainable AI, and interactive visualization within a unified software platform.

Achievement of Approved Objectives

Table 5.1 Achievement of Project Objectives

Approved Objective	Work Accomplished
Develop an AI-powered geopolitical intelligence platform.	Successfully developed the Forsyt platform integrating AI, NLP, and financial analytics.
Collect and analyse geopolitical news.	Implemented automated news collection and NLP-based event extraction for structured geopolitical intelligence.
Develop an India-centric geopolitical risk indicator.	Designed and implemented the India AI-GPR Index for estimating geopolitical risk.
Analyse the impact of geopolitical events on Indian financial markets.	Developed the Indian Macroeconomic Intelligence module for analysing market indicators and macroeconomic trends.
Assess geopolitical risks affecting trade and supply chains.	Implemented the Trade & Supply Chain Intelligence module for monitoring strategic trade corridors and logistics risks.
Evaluate portfolio exposure to geopolitical developments.	Developed the Portfolio Exposure & GPR Analytics module to support investment risk assessment.
Improve transparency through Explainable AI.	Integrated explainable AI techniques to enhance the interpretability of analytical outputs.

Develop an interactive decision-support interface.	Designed and implemented a modern web-based interface for visualization and analysis.
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Major Deliverables

The project successfully delivered the following software components:

- FORSYT Landing Page
- News Intelligence Module
- Indian Macroeconomic Intelligence Module
- Trade & Supply Chain Intelligence Module
- Portfolio Exposure & GPR Analytics Module
- India AI-GPR Index Generation
- NLP-Based Event Extraction Pipeline
- Interactive Dashboard and Visualizations

Overall Outcome

The completed Forsyt platform successfully satisfies the approved project objectives by integrating geopolitical intelligence, macroeconomic analysis, trade and supply chain monitoring, portfolio risk analytics, and explainable AI into a single decision-support platform. The modular architecture, interactive user interface, and AI-driven analytical capabilities provide a scalable foundation for supporting geopolitical risk assessment within the Indian financial ecosystem.

5.2 Conclusions

5.2 Conclusions

The successful development of **Forsyt** demonstrates the feasibility of integrating Artificial Intelligence, Natural Language Processing, geopolitical intelligence, and financial analytics into a unified decision-support platform for the Indian financial ecosystem. The project was undertaken to address the limitations of existing geopolitical monitoring systems by providing an India-centric solution capable of analysing geopolitical events and evaluating their potential impact on financial markets, trade corridors, supply chains, and investment portfolios.

Throughout the project, the proposed objectives were successfully achieved by implementing an end-to-end analytical platform that automates the collection of geopolitical news, performs NLP-based event extraction, generates the India AI-GPR Index, and integrates macroeconomic intelligence, trade and supply chain analysis, portfolio exposure assessment, and explainable AI within a single application. The modular architecture adopted during development ensured that individual analytical components could operate efficiently while contributing to an integrated and scalable software solution.

The project also demonstrates the practical application of modern AI technologies in solving real-world analytical problems. By combining automated news intelligence, predictive analytics, and interactive visualization, Forsyt transforms large volumes of unstructured geopolitical information into structured insights that can support informed decision-making. The incorporation of Explainable AI further enhances the transparency and interpretability of analytical outputs, improving user confidence in the generated results.

Overall, Forsyt successfully establishes a comprehensive geopolitical intelligence platform tailored to the Indian financial ecosystem. The completed system provides a strong foundation for future research and development in AI-driven geopolitical risk assessment and demonstrates how emerging technologies can be effectively applied to support strategic financial analysis, investment decision-making, and supply chain risk management in an increasingly interconnected global environment.

5.3 Environmental Benefits

5.3 Environmental, Economic, and Social Benefits

The implementation of **Forsyt** extends beyond technological innovation by providing benefits across environmental, economic, and social dimensions. By integrating Artificial Intelligence, Natural Language Processing, and geopolitical intelligence into a unified analytical platform, the system supports informed decision-making while promoting sustainable financial and operational practices.

5.3.1 Environmental Benefits

Forsyt contributes to environmental sustainability by supporting data-driven decision-making that minimizes operational disruptions and resource wastage.

Key environmental benefits include:

- Reduces the need for manual monitoring of multiple geopolitical news sources through automated intelligence gathering.
- Supports efficient supply chain planning by identifying geopolitical risks affecting major trade corridors.
- Enables proactive logistics planning, helping organizations reduce unnecessary transportation disruptions and associated environmental impacts.
- Promotes digital monitoring and reporting, reducing dependence on paper-based documentation and manual reporting processes.
- Encourages efficient utilization of resources through early identification of geopolitical uncertainties.

5.3.2 Economic Benefits

Forsyt provides valuable analytical support for financial institutions, businesses, investors, and policymakers by enabling informed economic decision-making.

Key economic benefits include:

- Supports investment decisions through geopolitical risk assessment.
- Assists in analysing the impact of geopolitical developments on Indian financial markets.
- Improves portfolio risk management by identifying geopolitical exposure.
- Enhances supply chain resilience by monitoring strategic trade routes.
- Provides early geopolitical intelligence that supports proactive business planning.

- Reduces analytical effort by automating data collection, processing, and visualization.

5.3.3 Social Benefits

The platform also contributes to broader societal benefits by improving access to reliable geopolitical intelligence and promoting informed decision-making.

Key social benefits include:

- Improves public understanding of global geopolitical developments through structured analytical insights.
- Supports researchers and students by providing a centralized geopolitical intelligence platform.
- Assists policymakers in analysing the potential impact of geopolitical events on national interests.
- Promotes transparency through Explainable AI, enabling users to understand the reasoning behind analytical outputs.
- Encourages evidence-based decision-making by integrating multiple sources of geopolitical and financial information.
- Enhances awareness of geopolitical risks affecting India's economy and strategic interests.

Summary

Forsyt delivers benefits across environmental, economic, and social dimensions by combining geopolitical intelligence with Artificial Intelligence and financial analytics. The platform enables informed decision-making, improves operational efficiency, strengthens financial risk assessment, and promotes greater awareness of geopolitical developments affecting the Indian financial ecosystem.

5.4 Future Work Plan

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Although Forsyt successfully demonstrates the application of Artificial Intelligence for geopolitical intelligence and financial analytics, several enhancements can further improve the platform's capabilities. The following future developments are proposed to enhance scalability, analytical accuracy, and practical usability.

Proposed Future Enhancements

1. Real-Time Data Streaming

Integrate real-time news streaming and financial market data to enable continuous monitoring of geopolitical developments and immediate updates to analytical results.

2. Expanded Geopolitical Coverage

Extend the platform to analyse geopolitical developments across additional countries and regions while maintaining dedicated India-centric analytical capabilities.

3. Enhanced Predictive Analytics

Incorporate advanced forecasting models capable of analysing long-term geopolitical trends and their potential influence on financial markets and economic indicators.

4. Advanced Explainable AI

Enhance the explainability module by providing more detailed visual explanations and interactive analytical reports to improve user understanding of model predictions.

5. Mobile Platform Support

Develop mobile applications to enable users to access geopolitical intelligence, risk assessments, and financial analytics through smartphones and tablets.

6. Additional Financial Indicators

Expand the analytical framework by incorporating additional economic indicators, sector-specific metrics, commodity prices, and other financial datasets to improve decision support.

7. Personalized Decision Support

Implement customizable dashboards and user-specific analytical recommendations based on individual investment preferences and organizational requirements.

8. Continuous Model Improvement

Periodically retrain and validate AI models using newly available geopolitical and financial data to improve analytical accuracy and adaptability to evolving global events.

Summary

The proposed future enhancements aim to improve the scalability, intelligence, and practical applicability of Forsyt. By incorporating real-time analytics, broader geopolitical coverage, enhanced predictive capabilities, mobile accessibility, and continuous model improvement, the platform can evolve into a more comprehensive decision-support system for geopolitical risk analysis and financial intelligence.

REFERENCES

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